MIXING SOLUTIONS FOR UNSTABLE MUSKAT DATA WITH VISCOSITY JUMP

INDEPENDENT RECONSTRUCTION FROM A SEMINAR ABSTRACT

Abstract. We consider the incompressible porous media (IPM) equation for two fluids with different densities and different viscosities, in the regime where the interface is fully unstable in the sense of Saffman and Taylor, so that the classical Muskat problem is ill-posed. For every horizontally periodic initial interface of class $C^{6,\gamma}$ whose Saffman–Taylor function (the jump of the normal derivative of the pressure) is negative everywhere, we construct infinitely many *mixing solutions*: weak solutions of IPM with viscosity jump whose density takes only the two pure values, coincides with the pure phases outside a strip, and mixes at arbitrarily small scales inside it. The strip opens linearly in time around an evolving central interface, at a local rate that is bounded by the Saffman–Taylor function. The construction combines the h-principle for IPM with viscosity jump with piecewise constant subsolutions defined through a second order expansion in time. The key ingredient is to write the Darcy field $\mu u = -\nabla p - \rho e_2$ of a subsolution as a sum of Cauchy integrals whose densities depend linearly on the Saffman–Taylor function, and to invert the resulting operator $I - 2A\mathcal{K}$, where A is the viscosity Atwood number and $\mathcal{K}$ is the adjoint Neumann–Poincaré operator of the interface. Thus the Birkhoff–Rott operator is inverted in terms of the Saffman–Taylor function rather than the vorticity, and at every order of the expansion one only has to invert the same Fredholm operator, which is a compact perturbation of the identity.

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1. Introduction

1.1. **The equations.** We study the motion of two incompressible fluids in a two-dimensional, horizontally periodic porous medium (or Hele-Shaw cell) under gravity. The fluids have different densities and different viscosities. Their evolution is governed by the transport equation for the density together with Darcy's law, the *incompressible porous media equation with viscosity*

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Provenance. This manuscript was written from the one-paragraph abstract of the talk *"Mixing solutions for unstable Muskat data with viscosity jump"* by Paula Luna (Universidad de Sevilla), joint work with Francisco Mengual, announced for the Seminario EDPs y Mecánica de Fluidos (ICMAT, 12 November 2026). It was prepared *without* access to their manuscript, as an exercise in reconstructing a research paper from an abstract. The result announced in that abstract is due to Luna and Mengual; the arguments, normalisations and any errors below are ours. This text is not intended for submission as original work.

jump:

$$\begin{cases} \partial_t \rho + \operatorname{div}(\rho u) = 0, \\ \operatorname{div} u = 0, \\ \mu(\rho)\, u = -\nabla p - \rho\, e_2, \end{cases} \qquad (x,t) \in \mathbb{T} \times \mathbb{R} \times (0,T), \tag{1.1}$$

where $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, $x = (x_1, x_2)$, $e_2 = (0,1)$, u is the velocity, p the pressure, ρ the density and $\mu(\rho)$ the viscosity. We have normalised the permeability and the gravity constant to one. Two-phase flows correspond to ρ taking only two values. After subtracting a hydrostatic pressure and rescaling space, time and pressure, we may assume that

$$\rho \in \{-1, 1\}, \qquad \mu(\rho) = 1 + A\rho, \qquad A = \frac{\mu^+ - \mu^-}{\mu^+ + \mu^-} \in (-1, 1), \tag{1.2}$$

where $\mu^+ = 1 + A$ is the viscosity of the denser fluid ($\rho = 1$) and $\mu^- = 1 - A$ that of the lighter one ($\rho = -1$); A is the viscosity Atwood number. The case $A = 0$ is the usual IPM equation. Since μ is affine in ρ and $\rho^2 = 1$, the momentum $m = \rho u$ allows to write Darcy's law *linearly* as $u + Am = -\nabla p - \rho e_2$; this observation, which goes back to [Men22], is the starting point of the convex integration theory for (1.1).

When the density is $\rho = 1$ above a curve $\Gamma(t)$ and $\rho = -1$ below it, (1.1) reduces to a free boundary problem for $\Gamma(t)$, the *Muskat problem* [Mus34]. Its well-posedness is governed by the sign of the jump of the normal derivative of the pressure across the interface,

$$\sigma := \partial_n p^+ - \partial_n p^- \qquad \text{on } \Gamma(t), \tag{1.3}$$

where n is the upward unit normal and $p^\pm$ are the traces of the pressure from above and below. We call σ the *Saffman–Taylor function*. If $\sigma > 0$ (the Rayleigh–Taylor, or Saffman–Taylor, condition) the problem is locally well-posed in Sobolev spaces [Amb04, CG07, CCG11, Mat19, NP20]; see the surveys [Gan17, GBL20]. If $\sigma < 0$ the problem is ill-posed in the sense of Hadamard [SCH04, CG07]: this is the Saffman–Taylor instability [ST58, Hom87], responsible for the fingering patterns observed when a less viscous fluid displaces a more viscous one, or when a heavier fluid lies on top of a lighter one. In our normalisation (see Proposition 2.7)

$$\sigma = -2\,(n_2 + A\,U), \tag{1.4}$$

where $U = u \cdot n$ is the normal velocity of the interface. The term $-2n_2$ is the gravitational (Rayleigh–Taylor) contribution and the term $-2AU$ the viscous (Saffman–Taylor) one: an interface is destabilised by gravity where the heavy fluid is on top ($n_2 > 0$), and by viscosity where the less viscous fluid pushes into the more viscous one ($AU > 0$). We say that Muskat initial data are *fully unstable* if $\sigma < 0$ along the whole initial interface.

1.2. **Mixing solutions.** In the unstable regime one does not expect classical solutions, and one is led to look for *weak* solutions which describe the formation of a *mixing zone*, a region where the two fluids mix at small scales. Otto [Ott99] proposed a relaxation of (1.1) (for flat interfaces) based on its gradient-flow structure, and showed that the mixing zone grows linearly in time. Using the convex integration method of De Lellis and Székelyhidi [DLS09] (see also [KMS03]), Córdoba, Faraco and Gancedo [CFG11] proved non-uniqueness of weak solutions of IPM (see also [Shv11]), and Székelyhidi [Sze12] computed the relaxation of IPM with $A = 0$ and constructed *mixing solutions* for the flat unstable interface, recovering Otto's linear growth. For non-flat interfaces with $A = 0$, Castro, Córdoba and Faraco [CCF21] constructed mixing solutions for every fully unstable H^5 interface; the mixing zone grows linearly around an evolving *pseudo-interface* whose evolution is governed by a delicate nonlocal equation, analysed with a semiclassical pseudodifferential calculus (see also [ACF21]). Förster and Székelyhidi [FS18] found a much shorter route via *piecewise constant* subsolutions which only involve the velocity and the acceleration of the interface at the initial time; Noisette and Székelyhidi [NS21] allowed variable opening speeds. Further developments include degraded mixing solutions [CFM19], localised mixing zones for bubbles and turned interfaces in the partially unstable regime [CFM22], and related results for the Rayleigh–Taylor instability and for vortex sheets [GKS21, MS23]. All these works concern equal viscosities, $A = 0$.

The case of different viscosities was addressed by Mengual [Men22], who proved an h-principle for (1.1) with $|A| < 1$ (weak solutions exist whenever a *subsolution* exists), computed the relevant relaxation, constructed mixing solutions for the *flat* unstable interface, and observed that the relaxation has a "pinch" singularity which prevents mixing where there is neither Rayleigh–Taylor instability nor vorticity at the interface.

The purpose of this paper is to construct mixing solutions with viscosity jump for *non-flat*, fully unstable Muskat data. Such a result was announced by Luna and Mengual [LM26]; the present text is an independent reconstruction written from the abstract of that announcement (see the note on the first page). The main difficulty with respect to [CCF21, FS18, NS21] is that, when $A \neq 0$, the velocity is no longer an explicit Birkhoff–Rott integral of the density: already for the sharp interface the vorticity is only implicitly determined by an integral equation (see (1.10) below), and for subsolutions the situation is worse, because the vorticity of u involves the curl of the (unknown) relaxed momentum.

1.3. **Main result.** Let $\mathcal{Z}^{k,\gamma}$ denote the set of horizontally periodic, non-self-intersecting $C^{k,\gamma}$ curves $z : \mathbb{R} \to \mathbb{R}^2$ with $z(\alpha) - (\alpha, 0)$ 2π-periodic (Definition 2.1); n_0 is the upward unit normal of $z_0 \in \mathcal{Z}^{k,\gamma}$ and $\Omega^{\pm}(z_0)$ the regions above and below z_0. The Muskat data are

$$\rho_0 = \mathbf{1}_{\Omega^+(z_0)} - \mathbf{1}_{\Omega^-(z_0)}. \tag{1.5}$$

In Proposition 2.7 we show that the Saffman–Taylor function σ_0 and the normal velocity U_0 of the Muskat problem at $t = 0$ are given by

$$\sigma_0 = -2\big(I - 2A\mathcal{K}(z_0)\big)^{-1}[n_{0,2}], \qquad U_0 = -\mathcal{K}(z_0)\sigma_0, \tag{1.6}$$

where $\mathcal{K}(z_0)$ is the adjoint Neumann–Poincaré operator of the curve (Definition 2.4). A weak solution of (1.1) is a pair $(\rho, u) \in L^\infty(\mathbb{T} \times \mathbb{R} \times (0,T))^{1+2}$ with $\rho^2 = 1$ a.e. such that the three equations in (1.1) hold in the sense of distributions on $\mathbb{T} \times \mathbb{R} \times [0,T)$ with initial datum ρ_0, where in the third one $\mu(\rho)u = u + A\rho u$ and the pressure is eliminated by taking the curl (see (3.1)).

Theorem 1.1. *Let $A \in (-1,1)$, $\gamma \in (0,1)$ and $z_0 \in \mathcal{Z}^{6,\gamma}$. Assume that the Muskat data (1.5) are fully unstable:*

$$\sigma_0(\alpha) < 0 \qquad \text{for every } \alpha \in \mathbb{T}. \tag{1.7}$$

Fix $\rho_{\mathrm{m}} \in (-1,1)$ and a function $\kappa \in C^{4,\gamma}(\mathbb{T})$ with

$$0 < \kappa(\alpha) < \frac{-\sigma_0(\alpha)}{2(1 - A\rho_{\mathrm{m}})} \qquad \text{for every } \alpha \in \mathbb{T}. \tag{1.8}$$

Then there exist $T > 0$ and curves $Z(\cdot,t) \in \mathcal{Z}^{3,\gamma}$, $t \in [0,T]$, of the form

$$Z(\alpha,t) = z_0(\alpha) + \Big(tU_0(\alpha) + \tfrac{t^2}{2}\, a(\alpha)\Big) n_0(\alpha) + h(t)\, e_2, \qquad a \in C^{3,\gamma}(\mathbb{T}), \quad |h(t)| \le Ct^3, \tag{1.9}$$

with the following property. Let

$$z^{\pm}(\alpha,t) := Z(\alpha,t) \pm t\,(1 \pm \rho_{\mathrm{m}})\,\kappa(\alpha)\, n_0(\alpha),$$

let $\Gamma^{\pm}(t)$ be the curves parametrised by $z^{\pm}(\cdot,t)$ and let $\Omega_{\mathrm{mix}}(t)$ be the open strip between them. Then there exist infinitely many weak solutions (ρ,u) of (1.1) on $[0,T]$ with initial datum ρ_0 such that

(i) *$\rho(x,t) = 1$ for a.e. x above $\Gamma^+(t)$ and $\rho(x,t) = -1$ for a.e. x below $\Gamma^-(t)$;*
(ii) *ρ mixes in Ω_{mix}: for every open set $B \subset \{(x,t) : 0 < t < T,\ x \in \Omega_{\mathrm{mix}}(t)\}$ both $\{\rho = 1\} \cap B$ and $\{\rho = -1\} \cap B$ have positive measure;*
(iii) *the coarse-grained density in Ω_{mix} is ρ_{m}: there is a sequence of such solutions converging weakly-$*$ in L^∞ to the function equal to 1 above $\Gamma^+(t)$, ρ_{m} in $\Omega_{\mathrm{mix}}(t)$, and -1 below $\Gamma^-(t)$.*

Thus the mixing zone has width $2t\kappa(\alpha)$ in the direction $n_0(\alpha)$; it grows linearly in time around the evolving central interface $Z(\cdot,t)$, which starts with the normal velocity U_0 of the Muskat problem, and its upper and lower boundaries move away from Z with normal speeds $(1+\rho_{\mathrm{m}})\kappa$ and $(1-\rho_{\mathrm{m}})\kappa$. By (1.8) the opening rate 2κ is bounded by $-\sigma_0/(1-A\rho_{\mathrm{m}})$, i.e. by the Saffman–Taylor function: the more unstable the interface, the faster the mixing zone may open. Note that the central interface Z is the "centre of mass" of the mixing zone, see Remark 4.4.

Remark 1.2. (a) For $A=0$ one has $\sigma_0=-2n_{0,2}$, so (1.7) says that z_0 is a graph with the heavy fluid on top, and Theorem 1.1 reduces to a result of the type of [CCF21, FS18, NS21] (with more regularity than needed there). For $A\neq 0$ condition (1.7) is not a priori a graph condition, since the viscous term $-2AU_0$ in (1.4) may compensate $n_{0,2}$.

(b) The admissible range (1.8) is the one allowed by a subsolution with a single intermediate density ρ_{m}. In Section 6 we show (formally) that subsolutions with continuous profiles allow the upper and lower boundaries to move away from the central interface with normal speeds arbitrarily close to $-\sigma_0/(1-A)$ and $-\sigma_0/(1+A)$ respectively (with a common factor $\theta<1$); for the flat interface ($\sigma_0\equiv -2$) this gives the asymmetric speeds $2/(1-A)$ and $2/(1+A)$ of Example 3.6.

(c) We have not tried to optimise the regularity of z_0.

1.4. **The key idea: the Saffman–Taylor function as unknown.** Let us explain the main new ingredient. In the classical treatment of the Muskat problem with viscosity jump [CCG11] one writes the velocity as the Birkhoff–Rott integral of its vorticity $\omega=\gamma\,\delta_\Gamma$, $\gamma=[\![u\cdot\tau]\!]$. Darcy's law and the continuity of the pressure then give the implicit equation

$$\big(I+2A\mathcal{K}(z)\big)\gamma=-2\tau_2, \tag{1.10}$$

see Remark 2.9. We proceed differently and work with the *Darcy field*

$$v:=\mu(\rho)\,u=-\nabla p-\rho e_2.$$

Its jump across the interface is $[\![v]\!]=-[\![\nabla p]\!]-[\![\rho]\!]e_2=-\sigma n-2e_2$, because the pressure is continuous. Hence v is the Cauchy integral of the density $-\sigma n-2e_2$, which is *linear in the Saffman–Taylor function*, and the continuity of $u\cdot n=v\cdot n/\mu$ becomes the equation $(I-2A\mathcal{K}(z))\sigma=-2n_2$ in (1.6). In other words, we invert the Birkhoff–Rott operator in terms of σ instead of γ.

The advantage becomes decisive for subsolutions. For a relaxed state (ρ,u,m) the Darcy field $v=u+Am$ satisfies

$$\operatorname{curl} v=-\partial_1\rho,\qquad \operatorname{div} v=-A\,\partial_t\rho,$$

so v is *explicitly* determined by ρ and $\partial_t\rho$, whereas the vorticity of u is not (it involves $\operatorname{curl} m$). For a piecewise constant subsolution with two boundary curves $z^\pm$ moving with normal velocities $V^\pm$, the field v is the sum of two Cauchy integrals with densities $[\![\rho]\!]^\pm\big(-e_2+(n_2^\pm+AV^\pm)n^\pm\big)$, and the relaxation conditions only involve v, the normal velocities and a divergence-free field in the strip. When the strip collapses, the relevant combination of the densities is precisely $-\sigma n-2e_2$, the opening rate is controlled by σ, and the condition that the centre of mass of the mixing zone moves with the fluid is, at every order of the time expansion, an equation of the form $(I-2A\mathcal{K}(z_0))X=Y$. Since $\mathcal{K}(z_0)$ is a smoothing operator, $I-2A\mathcal{K}(z_0)$ is a compact perturbation of the identity, and its invertibility follows from a short energy argument.

1.5. **Strategy.** The proof has four steps.

Step 1. *Relaxation* (Section 3). In the variables (ρ,v,m) we compute the first and second order laminates of the constitutive set and obtain the open set

$$\mathcal{U}_A=\big\{(\rho,u,m):\ |\rho|<1,\ |D-a|<|a|\big\},\qquad D=\tfrac{2[(1-A^2)m-(\rho-A)v]}{1-\rho^2}+2e_2,\quad a=e_2+\tfrac{A}{1+A\rho}\,v,$$

which is contained in the interior of the Λ-convex hull. Subsolutions with values in $\mathcal{U}_A$ yield mixing solutions by the h-principle of [Men22].

Step 2. *Ansatz* (Section 4). Given two curves $z^{\pm}$ bounding a strip with density ρ_{m}, we define v as a sum of Cauchy integrals, the relaxed momentum outside the strip by the constitutive relation, and inside the strip through a stream function interpolating the fluxes prescribed by the Rankine–Hugoniot conditions. The linear equations then hold *exactly*, for any curves which conserve the total mass. The only remaining condition is the open inequality $|D-a|<|a|$, and it is controlled by a scalar *flux mismatch* $M(\alpha,t)$.

Step 3. *Expansion in time* (Section 5). Following [FS18], we take curves which are second order polynomials in t. The vanishing of M at order t^0 is the Muskat equation (1.6) for the normal velocity; at order t^1 it is an equation $(I-2A\mathcal{K}(z_0))a=R_1$ for the normal acceleration a. A small vertical translation restores exact mass conservation, and then $M=O(t^2)$.

Step 4. *Conclusion.* With $M=O(t^2)$ the state in the strip is $O(t^\gamma)$-close to its limit as $t\to 0$, which lies in $\mathcal{U}_A$ exactly when (1.8) holds.

Organisation. Section 2 introduces periodic Cauchy integrals, the operator $\mathcal{K}(z)$ and the Saffman–Taylor formulation of the Muskat problem. Section 3 contains the relaxation and the h-principle, together with the explicit flat case. Section 4 introduces the piecewise constant ansatz and Section 5 proves Theorem 1.1. Section 6 contains remarks on optimal opening rates and open problems. The appendix collects the properties of Cauchy integrals that we use.

2. Cauchy integrals and the Saffman–Taylor formulation of the Muskat problem

2.1. **Curves and notation.** We identify $\mathbb{R}^2$ with $\mathbb{C}$, $x=x_1+ix_2$. For a vector $w=(w_1,w_2)$ we write $w^\flat := w_1-iw_2$ (the complex conjugate of w seen as a complex number); a vector field w is divergence-free and curl-free in an open set if and only if $w^\flat$ is holomorphic there. We write $w^\perp=(-w_2,w_1)$.

Definition 2.1. For $k\geq 1$ and $\gamma\in(0,1)$ let $\mathcal{Z}^{k,\gamma}$ be the set of maps $z\in C^{k,\gamma}_{\mathrm{loc}}(\mathbb{R};\mathbb{R}^2)$ such that $z(\alpha)-(\alpha,0)$ is 2π-periodic and

$$\delta(z) := \inf\left\{ \frac{|\sin\frac{z(\alpha)-z(\beta)}{2}|}{|\sin\frac{\alpha-\beta}{2}|} :\ \alpha,\beta\in\mathbb{R},\ \alpha-\beta\notin 2\pi\mathbb{Z}\right\} > 0,$$

where sin is the complex sine. We norm $\mathcal{Z}^{k,\gamma}$ by $\|z-(\cdot,0)\|_{C^{k,\gamma}(\mathbb{T})}$.

Since $|\sin\frac{w}{2}|^2=\sin^2\frac{\operatorname{Re}w}{2}+\sinh^2\frac{\operatorname{Im}w}{2}$, the condition $\delta(z)>0$ says that z induces an embedding of $\mathbb{T}$ into the cylinder $\mathbb{T}\times\mathbb{R}$, quantitatively; in particular $|\partial_\alpha z|\geq\delta(z)$. The curve $\Gamma=z(\mathbb{R})$ is invariant under $x\mapsto x+2\pi e_1$ and $\mathbb{R}^2\setminus\Gamma$ has two connected components $\Omega^+(z)$ and $\Omega^-(z)$, containing $\{x_2>\max z_2\}$ and $\{x_2<\min z_2\}$ respectively. We write

$$\tau=\frac{\partial_\alpha z}{|\partial_\alpha z|},\qquad n=\tau^\perp\ \ (\text{so } n=i\tau \text{ as complex numbers}),\qquad \mathrm{d}s=|\partial_\alpha z|\,\mathrm{d}\alpha,$$

so that n points into $\Omega^+(z)$, and $n_2=\partial_\alpha z_1/|\partial_\alpha z|$. For a function w defined near Γ we denote by $w^\pm(\alpha)$ its limits at $z(\alpha)$ from $\Omega^\pm(z)$, by $[\![w]\!]=w^+-w^-$ its jump and by $\langle w\rangle=\frac12(w^++w^-)$ its average. The constant C may change from line to line.

2.2. **Periodic Cauchy integrals.**

Definition 2.2. Let $z\in\mathcal{Z}^{1,\gamma}$ and $f\in C^{0,\gamma}(\mathbb{T};\mathbb{R}^2)$. The *Cauchy field* of f along z is the vector field $\mathcal{C}[z;f]$ on $\mathbb{R}^2\setminus\Gamma$ defined by

$$\mathcal{C}[z;f]^\flat(x)=\frac{1}{4\pi i}\int_{\mathbb{T}} f^\flat(\beta)\,\partial_\beta z(\beta)\,\cot\frac{z(\beta)-x}{2}\,\mathrm{d}\beta. \tag{2.1}$$

Since $\frac{1}{2}\cot\frac{w}{2} = \frac{1}{w} + O(w)$, formula (2.1) is the periodisation of the classical Cauchy integral

$$\frac{1}{2\pi i}\int \frac{f^\flat(\zeta)\,\mathrm{d}\zeta}{\zeta - x}.$$

Lemma 2.3. *Let $z \in \mathcal{Z}^{1,\gamma}$, $f \in C^{0,\gamma}(\mathbb{T};\mathbb{R}^2)$ and $\mathcal{C} = \mathcal{C}[z;f]$.*

(a) *$\mathcal{C}$ is real analytic and 2π-periodic in x_1 on $\mathbb{R}^2 \setminus \Gamma$, with $\operatorname{div}\mathcal{C} = \operatorname{curl}\mathcal{C} = 0$ there.*

(b) (Plemelj–Privalov) *The restrictions of $\mathcal{C}$ to $\Omega^\pm(z)$ extend to $C^{0,\gamma}(\overline{\Omega^\pm(z)})$ with*

$$\|\mathcal{C}\|_{C^{0,\gamma}(\overline{\Omega^\pm})} \le C\|f\|_{C^{0,\gamma}},$$

where C depends on γ, $\delta(z)$ and $\|z\|_{\mathcal{Z}^{1,\gamma}}$. Moreover $[\![\mathcal{C}]\!] = f$, and $\langle\mathcal{C}\rangle$ is given by (2.1) with $x = z(\alpha)$ and the integral understood in the principal value sense.

(c) *As $x_2 \to \pm\infty$, $\mathcal{C}(x) \to \mathcal{C}_{\pm\infty}$ uniformly in x_1, where $\mathcal{C}^\flat_{\pm\infty} = \pm\frac{1}{4\pi}\int_{\mathbb{T}} f^\flat \partial_\beta z\,\mathrm{d}\beta$. In particular, for a constant vector c we have $\mathcal{C}[z;c] = \pm\frac{c}{2}$ in $\Omega^\pm(z)$, and for a scalar ϕ*

$$\mathcal{C}[z;\phi n]_{\pm\infty} = \pm\frac{1}{4\pi}\Big(\int_{\mathbb{T}} \phi\,|\partial_\beta z|\,\mathrm{d}\beta\Big)e_2. \tag{2.2}$$

(d) *If $j \ge 0$, $z \in \mathcal{Z}^{j+1,\gamma}$ and $f \in C^{j,\gamma}$, then $\mathcal{C} \in C^{j,\gamma}(\overline{\Omega^\pm(z)})$ and $\mathcal{C}^\pm \in C^{j,\gamma}(\mathbb{T})$, with bounds.*

(e) (Representation) *Let w be a bounded vector field on $\mathbb{R}^2\setminus\Gamma$, 2π-periodic in x_1, divergence-free and curl-free in $\mathbb{R}^2 \setminus \Gamma$, whose restrictions to $\Omega^\pm(z)$ extend continuously to the closures with $[\![w]\!] \in C^{0,\gamma}(\mathbb{T})$, and with $w(x) \to 0$ as $x_2 \to \pm\infty$. Then $w = \mathcal{C}[z;[\![w]\!]]$.*

Proof. (a) is immediate. (b) and (d) are the periodic versions of the Plemelj–Privalov theorem and are discussed in Appendix A. (c) When $x_2 \to +\infty$ (resp. $-\infty$), $\cot\frac{z(\beta)-x}{2} \to i$ (resp. $-i$) uniformly in β, which gives $\mathcal{C}_{\pm\infty}$. If $f = c$ is constant, then $\partial_x\int_{\mathbb{T}} c^\flat\partial_\beta z\cot\frac{z(\beta)-x}{2}\,\mathrm{d}\beta = -c^\flat\int_{\mathbb{T}}\partial_\beta\big[\cot\frac{z(\beta)-x}{2}\big]\,\mathrm{d}\beta = 0$ by periodicity, so $\mathcal{C}[z;c]$ is constant in each component and equals its limit $\pm\frac{1}{4\pi}c^\flat\cdot 2\pi = \pm\frac{c^\flat}{2}$; recall $\int_{\mathbb{T}}\partial_\beta z\,\mathrm{d}\beta = 2\pi$. For $f = \phi n$ we have $n^\flat\partial_\beta z = -i|\partial_\beta z|$, so $\mathcal{C}^\flat_{+\infty} = -\frac{i}{4\pi}\int\phi|\partial_\beta z|$, i.e. (2.2). (e) The field $W = w - \mathcal{C}[z;[\![w]\!]]$ is bounded, periodic, continuous across Γ, and $W^\flat$ is holomorphic off Γ; hence $W^\flat$ is entire (Morera), and by Liouville's theorem on the cylinder it is constant. By (c), $W \to -\mathcal{C}_{+\infty}$ as $x_2 \to +\infty$ and $W \to -\mathcal{C}_{-\infty} = \mathcal{C}_{+\infty}$ as $x_2 \to -\infty$. Therefore $\mathcal{C}_{+\infty} = 0$ and $W \equiv 0$. □

2.3. **The adjoint Neumann–Poincaré operator.**

Definition 2.4. For $z \in \mathcal{Z}^{2,\gamma}$ and a real function $\phi \in C^{0,\gamma}(\mathbb{T})$ we set

$$\mathcal{K}(z)\phi := \langle\mathcal{C}[z;\phi n]\rangle\cdot n.$$

Since the jump of $\mathcal{C}[z;\phi n]$ is normal, Lemma 2.3(b) gives

$$\mathcal{C}[z;\phi n]^\pm\cdot n = \mathcal{K}(z)\phi \pm \tfrac{1}{2}\phi, \qquad \mathcal{C}[z;\phi n]^+\cdot\tau = \mathcal{C}[z;\phi n]^-\cdot\tau. \tag{2.3}$$

In potential-theoretic terms, $\mathcal{C}[z;\phi n]$ is the gradient of the single layer potential of ϕ and $\mathcal{K}(z)$ is the (periodic) adjoint Neumann–Poincaré operator.

Lemma 2.5. *Let $k \ge 2$ and $z \in \mathcal{Z}^{k,\gamma}$.*

(a) *$\mathcal{K}(z)\phi(\alpha) = \int_{\mathbb{T}} k_z(\alpha,\beta)\phi(\beta)\,\mathrm{d}\beta$, with*

$$k_z(\alpha,\beta) = \frac{|\partial_\beta z(\beta)|}{4\pi|\partial_\alpha z(\alpha)|}\,\operatorname{Im}\Big[\partial_\alpha z(\alpha)\cot\frac{z(\beta)-z(\alpha)}{2}\Big],$$

and k_z extends from $\{\alpha \ne \beta\}$ to a function in $C^{k-2,\gamma}(\mathbb{T}\times\mathbb{T})$. Consequently $\mathcal{K}(z)$ maps $C^0(\mathbb{T})$ boundedly into $C^{k-2,\gamma}(\mathbb{T})$, and for $k \ge 3$ it is compact on $C^{j,\gamma}(\mathbb{T})$ for $0 \le j \le k-3$.

(b) *$\int_{\mathbb{T}}\big(\mathcal{K}(z)\phi\big)\,|\partial_\alpha z|\,\mathrm{d}\alpha = 0$ for every $\phi \in C^{0,\gamma}(\mathbb{T})$.*

Proof. (a) The complex density of ϕn is $\phi\, n^\flat \partial_\beta z = -i\phi|\partial_\beta z|$. Using $w\cdot n = \mathrm{Re}(w^\flat n)$ and $n = i\partial_\alpha z/|\partial_\alpha z|$, the principal value formula of Lemma 2.3(b) gives the stated kernel, once we show it is bounded (then no principal value is needed). Away from the diagonal $\{\alpha-\beta\in 2\pi\mathbb{Z}\}$ the kernel is $C^{k-1,\gamma}$, because $\delta(z)>0$. Near the diagonal write $\cot\frac{w}{2} = \frac{2}{w}+g(w)$ with g analytic for $|w|<2\pi$, and

$$z(\beta)-z(\alpha) = (\beta-\alpha)Z_1(\alpha,\beta), \qquad \partial_\alpha z(\alpha) - Z_1(\alpha,\beta) = -(\beta-\alpha)Z_2(\alpha,\beta),$$

with $Z_1 = \int_0^1 \partial_\alpha z(\alpha+s(\beta-\alpha))\,\mathrm{d}s \in C^{k-1,\gamma}$, $Z_2 = \int_0^1\int_0^1 s\,\partial_\alpha^2 z(\alpha+rs(\beta-\alpha))\,\mathrm{d}r\,\mathrm{d}s \in C^{k-2,\gamma}$ and $|Z_1|\geq c>0$. Then

$$\mathrm{Im}\Big[\frac{2\partial_\alpha z(\alpha)}{z(\beta)-z(\alpha)}\Big] = \frac{2}{\beta-\alpha}\,\mathrm{Im}\Big[\frac{\partial_\alpha z(\alpha)-Z_1}{Z_1}\Big] = -2\,\mathrm{Im}\Big[\frac{Z_2}{Z_1}\Big] \in C^{k-2,\gamma},$$

while $\partial_\alpha z(\alpha) g(z(\beta)-z(\alpha)) \in C^{k-1,\gamma}$. Differentiating under the integral sign gives $\mathcal{K}(z): C^0\to C^{k-2,\gamma}$, and compactness follows from the compact embedding $C^{k-2,\gamma}(\mathbb{T})\subset C^{j,\gamma}(\mathbb{T})$ for $j\leq k-3$.

(b) Let $w=\mathcal{C}[z;\phi n]$. Since w is divergence-free and periodic, its flux through Γ from above equals its flux through $\{x_2=L\}$ for every large L, and letting $L\to\infty$ we get from (2.2) $\int_{\mathbb{T}} w^+\cdot n\,\mathrm{d}s = 2\pi(\mathcal{C}_{+\infty})_2 = \frac12\int\phi\,\mathrm{d}s$. Similarly $\int_{\mathbb{T}} w^-\cdot n\,\mathrm{d}s = 2\pi(\mathcal{C}_{-\infty})_2 = -\frac12\int\phi\,\mathrm{d}s$. Adding, $\int\langle w\rangle\cdot n\,\mathrm{d}s = 0$. □

Proposition 2.6. *Let $|A|<1$, $k\geq 3$ and $z\in\mathcal{Z}^{k,\gamma}$. Then $I-2A\mathcal{K}(z)$ is an isomorphism of $C^{j,\gamma}(\mathbb{T})$ for $0\leq j\leq k-3$. Moreover, if $f\in C^{j,\gamma}(\mathbb{T})$ with $0\leq j\leq k-2$, the solution $\sigma\in C^{0,\gamma}(\mathbb{T})$ of $(I-2A\mathcal{K}(z))\sigma = f$ belongs to $C^{j,\gamma}(\mathbb{T})$, with $\|\sigma\|_{C^{j,\gamma}}\leq C\|f\|_{C^{j,\gamma}}$, and $\int\sigma\,\mathrm{d}s = \int f\,\mathrm{d}s$.*

Proof. By Lemma 2.5(a) and the Riesz–Schauder theory, $I-2A\mathcal{K}(z)$ is a Fredholm operator of index zero on $C^{j,\gamma}(\mathbb{T})$, $0\leq j\leq k-3$; so it suffices to prove injectivity on $C^{0,\gamma}$. Let $\phi\in C^{0,\gamma}$ with $\phi = 2A\mathcal{K}(z)\phi$. If $A=0$ there is nothing to prove, so let $A\neq 0$. By Lemma 2.5(b), $\int\phi\,\mathrm{d}s = 0$, so $w := \mathcal{C}[z;\phi n]$ tends to 0 as $x_2\to\pm\infty$, by (2.2); being the conjugate of a periodic holomorphic function, it does so exponentially fast. Since the jump of w is normal, w is curl-free in $\mathcal{D}'(\mathbb{R}^2)$, and since its circulation along horizontal lines tends to zero, $w=\nabla\Phi$ for a periodic, Lipschitz function Φ which is $C^{1,\gamma}$ up to Γ from both sides and bounded. By (2.3) and $\mathcal{K}(z)\phi = \phi/(2A)$,

$$\frac{w^+\cdot n}{1+A} = \frac{w^-\cdot n}{1-A} = \frac{\phi}{2A} =: g.$$

Integrating by parts on one period of $\Omega^\pm\cap\{|x_2|<L\}$ and letting $L\to\infty$ (the contributions of the horizontal boundaries vanish by the exponential decay, the lateral ones cancel by periodicity),

$$\frac{1}{1+A}\int_{\Omega^+}|w|^2 + \frac{1}{1-A}\int_{\Omega^-}|w|^2 = -\int_{\mathbb{T}}\Phi\,\frac{w^+\cdot n}{1+A}\,\mathrm{d}s + \int_{\mathbb{T}}\Phi\,\frac{w^-\cdot n}{1-A}\,\mathrm{d}s = 0,$$

where the integrals over $\Omega^\pm$ are over one period. Hence $w\equiv 0$ and $\phi = [\![w]\!]\cdot n = 0$.

For the regularity statement with $j=k-2$: the solution $\sigma\in C^{k-3,\gamma}$ satisfies $\sigma = f + 2A\mathcal{K}(z)\sigma$, and $\mathcal{K}(z)\sigma\in C^{k-2,\gamma}$ by Lemma 2.5(a). Finally $\int\sigma\,\mathrm{d}s = \int f\,\mathrm{d}s$ by Lemma 2.5(b). □

2.4. **The Muskat problem with viscosity jump.** Let $z\in\mathcal{Z}^{k,\gamma}$, $k\geq 3$, and $\rho=\pm1$ in $\Omega^\pm(z)$. The *instantaneous Darcy problem* consists in finding u and p, $C^{0,\gamma}$ and $C^{1,\gamma}$ up to Γ from both sides, such that

$$\mathrm{div}\,u = 0, \quad (1\pm A)\,u = -\nabla p \mp e_2 \ \text{ in } \Omega^\pm(z), \qquad [\![p]\!] = 0, \ \ [\![u\cdot n]\!] = 0 \ \text{ on } \Gamma, \tag{2.4}$$

with $u\to 0$ and $\nabla p+\rho e_2\to 0$ as $x_2\to\pm\infty$ (the fluids are at rest at infinity). We introduce the *Darcy field* and the *Saffman–Taylor function*

$$v := \mu(\rho)u = -\nabla p - \rho e_2, \qquad \sigma := [\![\partial_n p]\!] = \partial_n p^+ - \partial_n p^- \ \text{ on } \Gamma.$$

Proposition 2.7. *Let $|A|<1$, $k\geq 3$, $z\in\mathcal{Z}^{k,\gamma}$, and let $\sigma\in C^{k-2,\gamma}(\mathbb{T})$ be the solution of*

$$\big(I-2A\mathcal{K}(z)\big)\sigma=-2n_2. \tag{2.5}$$

Then (2.4) has a solution, unique up to an additive constant in p, given by

$$v=-\mathcal{C}[z;\sigma n]\mp e_2\quad\text{and}\quad u=\frac{v}{1\pm A}\qquad\text{in }\Omega^{\pm}(z). \tag{2.6}$$

Its Saffman–Taylor function is σ, the normal velocity $U:=u\cdot n$ on Γ is

$$U=-\mathcal{K}(z)\sigma, \tag{2.7}$$

and moreover $\sigma=-2(n_2+AU)$ and $\int_{\mathbb{T}}\sigma\,\mathrm{d}s=-4\pi$.

Proof. Define v and u by (2.6). By (2.3), $v^{\pm}\cdot n=-\mathcal{K}\sigma\mp\frac12\sigma\mp n_2$, and (2.5) says $\frac12\sigma+n_2=A\mathcal{K}\sigma$. Therefore $v^{\pm}\cdot n=-(1\pm A)\mathcal{K}\sigma$, i.e. $u^+\cdot n=u^-\cdot n=-\mathcal{K}\sigma$, which gives $[\![u\cdot n]\!]=0$ and (2.7); then $\sigma=-2n_2+2A\mathcal{K}\sigma=-2(n_2+AU)$. Integrating (2.5) and using Lemma 2.5(b) and $\int n_2\,\mathrm{d}s=\int\partial_\alpha z_1\,\mathrm{d}\alpha=2\pi$, we get $\int\sigma\,\mathrm{d}s=-4\pi$; by (2.2), $v\to\mp\frac{1}{4\pi}(\int\sigma\,\mathrm{d}s)e_2\mp e_2=0$ as $x_2\to\pm\infty$. Next, $v+\rho e_2=-\mathcal{C}[z;\sigma n]$ in $\mathbb{R}^2\setminus\Gamma$; this field is curl-free in $\mathcal{D}'(\mathbb{R}^2)$, because its jump is normal, and its circulation along horizontal lines vanishes (its limits at $\pm\infty$ are vertical). Hence $v+\rho e_2=-\nabla p$ for a continuous function p, periodic in x_1, and $\partial_n p^{\pm}=\mathcal{C}[z;\sigma n]^{\pm}\cdot n=\mathcal{K}\sigma\pm\frac12\sigma$, so that $[\![\partial_n p]\!]=\sigma$. The field u is divergence-free in each $\Omega^{\pm}$ and has no normal jump, so $\operatorname{div}u=0$ in $\mathcal{D}'(\mathbb{R}^2)$. Regularity follows from Lemma 2.3(d).

For uniqueness, let (u',p') be the difference of two solutions. Then $\operatorname{div}u'=0$, $(1\pm A)u'=-\nabla p'$ in $\Omega^{\pm}$, p' is continuous, $u'\cdot n$ has no jump, and $u'\to 0$ at infinity, exponentially fast since u' is the gradient of a periodic harmonic function in $\{|x_2|>L\}$. Integrating by parts over one period,

$$\int(1+A\rho)|u'|^2=-\int\nabla p'\cdot u'=\int_{\mathbb{T}}p'\left(u'^{+}-u'^{-}\right)\cdot n\,\mathrm{d}s=0,$$

so $u'=0$ and $\nabla p'=0$. □

Since the interface moves with the normal velocity of the fluid, Proposition 2.7 gives the following formulation of the Muskat problem with viscosity jump, in which the only operator to be inverted acts on the Saffman–Taylor function.

Corollary 2.8. *A family of curves $z(\cdot,t)\in\mathcal{Z}^{k,\gamma}$, $k\geq 3$, solves the Muskat problem with viscosity jump if and only if*

$$\partial_t z\cdot n=-\mathcal{K}(z)\sigma,\qquad\big(I-2A\mathcal{K}(z)\big)\sigma=-2n_2.$$

Remark 2.9 (Duality with the vorticity formulation). For a real function γ, the complex densities of $\gamma\tau$ and γn are $\gamma|\partial_\beta z|$ and $-i\gamma|\partial_\beta z|$, so $\mathcal{C}[z;\gamma\tau]^\flat=i\,\mathcal{C}[z;\gamma n]^\flat$, i.e. $\mathcal{C}[z;\gamma\tau]$ is obtained by rotating $\mathcal{C}[z;\gamma n]$ by $-\pi/2$. In particular $\langle\mathcal{C}[z;\gamma\tau]\rangle\cdot\tau=\langle\mathcal{C}[z;\gamma n]\rangle\cdot n=\mathcal{K}(z)\gamma$. The velocity is curl-free off Γ with jump $\gamma\tau$, $\gamma=[\![u\cdot\tau]\!]$, hence $u=\mathcal{C}[z;\gamma\tau]$ by Lemma 2.3(e): $\langle u\rangle$ is the Birkhoff–Rott integral of the vortex sheet. The tangential part of Darcy's law, $[\![v\cdot\tau]\!]=-[\![\rho]\!]\tau_2=-2\tau_2$, reads $\gamma+2A\langle u\rangle\cdot\tau=-2\tau_2$, i.e. (1.10); with $\varpi=\gamma|\partial_\alpha z|$ this is the familiar equation $\varpi+2A\,\mathrm{BR}(z,\varpi)\cdot\partial_\alpha z=-2\partial_\alpha z_2$ of [CCG11]. Thus the vorticity is obtained by inverting $I+2A\mathcal{K}(z)$, the Saffman–Taylor function by inverting $I-2A\mathcal{K}(z)$; both are covered by Proposition 2.6. Note however that in the vorticity formulation the normal velocity $U=\langle u\rangle\cdot n=-\langle\mathcal{C}[z;\gamma n]\rangle\cdot\tau$ is a singular integral of γ, whereas in (2.7) it is the smoothing operator $\mathcal{K}(z)$ applied to σ. More importantly, the Saffman–Taylor formulation extends to subsolutions, where the vorticity formulation does not (Section 4).

Definition 2.10. Let $z_0\in\mathcal{Z}^{k,\gamma}$, $k\geq 3$. The Saffman–Taylor function of the Muskat data (1.5) is $\sigma_0:=-2(I-2A\mathcal{K}(z_0))^{-1}[n_{0,2}]$, and the data are *fully unstable* if $\sigma_0<0$ on $\mathbb{T}$. We write $U_0:=-\mathcal{K}(z_0)\sigma_0$ and $\mathcal{K}_0:=\mathcal{K}(z_0)$.

Example 2.11. For the flat interface $z_0(\alpha)=(\alpha,0)$ the kernel k_{z_0} vanishes (the cotangent of a real number is real), so $\mathcal{K}_0=0$, $\sigma_0\equiv-2$ and $U_0\equiv 0$. If $A=0$, then $\sigma_0=-2n_{0,2}$ for every z_0, and the data are fully unstable if and only if $\partial_\alpha z_{0,1}>0$, i.e. z_0 is a graph with the heavier fluid on top.

For $z_0 \in \mathcal{Z}^{6,\gamma}$, Proposition 2.6 (with $k=6$, $j=4$) and Lemma 2.5(a) give $\sigma_0, U_0 \in C^{4,\gamma}(\mathbb{T})$.

3. Relaxation of IPM with viscosity jump

3.1. **IPM as a differential inclusion.** A pair $(\rho,u) \in L^\infty(\mathbb{T}\times\mathbb{R}\times(0,T))^{1+2}$ with $\rho^2=1$ a.e. is a *weak solution* of (1.1) with initial datum ρ_0 if for every $\varphi \in C_c^\infty(\mathbb{T}\times\mathbb{R}\times[0,T))$

$$\begin{aligned} &\int_0^T\!\!\int \big(\rho\,\partial_t\varphi + \rho u\cdot\nabla\varphi\big)\,\mathrm{d}x\,\mathrm{d}t + \int \rho_0\varphi(\cdot,0)\,\mathrm{d}x = 0, \qquad \int_0^T\!\!\int u\cdot\nabla\varphi\,\mathrm{d}x\,\mathrm{d}t = 0,\\ &\int_0^T\!\!\int \big((1+A\rho)\,u\cdot\nabla^\perp\varphi + \rho\,\partial_1\varphi\big)\,\mathrm{d}x\,\mathrm{d}t = 0, \end{aligned} \tag{3.1}$$

where $\nabla^\perp = (-\partial_2,\partial_1)$; the last identity is the curl of Darcy's law, $\operatorname{curl}(\mu(\rho)u) = -\partial_1\rho$. Consider the linear system for $(\rho,u,m)\in\mathbb{R}\times\mathbb{R}^2\times\mathbb{R}^2$

$$\partial_t\rho + \operatorname{div} m = 0, \qquad \operatorname{div} u = 0, \qquad \operatorname{curl}(u+Am) = -\partial_1\rho, \tag{3.2}$$

and the constitutive set

$$\mathsf{K} := \{(\rho,u,m):\ \rho\in\{-1,1\},\ m=\rho u\}.$$

Since $(1+A\rho)u = u+Am$ on K, a pair (ρ,u) is a weak solution if and only if $(\rho,u,\rho u)$ solves (3.2) in the sense of distributions, with initial datum ρ_0, and takes values in K a.e. This linearisation of Darcy's law is due to [Men22].

For a state (ρ,u,m) we define the *Darcy field* $v := u+Am$. In terms of (ρ,v,m) the system (3.2) becomes

$$\partial_t\rho + \operatorname{div} m = 0, \qquad \operatorname{curl} v = -\partial_1\rho, \qquad \operatorname{div} v = -A\,\partial_t\rho, \qquad u = v - Am. \tag{3.3}$$

The point of (3.3) is that *v is determined by ρ and $\partial_t\rho$ alone* (modulo harmonic fields), whereas the vorticity of $u = v - Am$ depends on $\operatorname{curl} m$. On K we have $v = (1+A\rho)u = \mu(\rho)u = -\nabla p - \rho e_2$.

3.2. **Laminates.** The wave cone of (3.2) is the set Λ of states $\bar W = (\bar\rho,\bar u,\bar m)$ for which there exist $\xi\in\mathbb{R}^2\setminus\{0\}$ and $c\in\mathbb{R}$ such that $\bar W h(\xi\cdot x + ct)$ solves (3.2) for every $h\in C^1(\mathbb{R})$, that is,

$$c\bar\rho + \xi\cdot\bar m = 0, \qquad \xi\cdot\bar u = 0, \qquad \xi^\perp\cdot(\bar u + A\bar m) + \xi_1\bar\rho = 0. \tag{3.4}$$

For a state (ρ,u,m) with $|\rho|<1$ we introduce

$$\mu := 1+A\rho, \quad v := u+Am, \quad d := \frac{2\big[(1-A^2)m - (\rho-A)v\big]}{1-\rho^2}, \quad D := d+2e_2, \quad a := e_2 + \frac{A}{\mu}v. \tag{3.5}$$

For fixed ρ the map $(u,m)\mapsto(v,d)$ is a linear isomorphism of $\mathbb{R}^4$, with inverse

$$m = \frac{(\rho-A)v + \frac12(1-\rho^2)d}{1-A^2}, \qquad u = v - Am.$$

Lemma 3.1 (First order laminates)**.** *Let $\rho\in(-1,1)$, $\lambda_\pm = \frac{1\pm\rho}{2}$, and $P_+ = (1,u_+,u_+)$, $P_- = (-1,u_-,-u_-)\in\mathsf{K}$. Then $P_+ - P_-\in\Lambda$ if and only if there is a unit vector ν with*

$$u_+\cdot\nu = u_-\cdot\nu, \qquad \big((1+A)u_+ - (1-A)u_-\big)\cdot\nu^\perp = -2\nu_1. \tag{3.6}$$

In that case the barycentre $P = \lambda_+P_+ + \lambda_-P_- = (\rho,u,m)$ satisfies

$$D = 2(a\cdot\nu)\nu, \qquad \text{in particular} \qquad |D-a| = |a|. \tag{3.7}$$

Conversely, for every $\rho\in(-1,1)$, $v\in\mathbb{R}^2$ and unit vector ν, the state with these values of ρ, v and with $D = 2(a\cdot\nu)\nu$ is such a barycentre.

Proof. We have $P_+ - P_- = (2, u_+-u_-, u_++u_-)$. Taking $\xi=\nu$ in (3.4), the first condition only fixes c, the second is $\nu\cdot(u_+-u_-)=0$ and the third is $\nu^\perp\cdot(u_+-u_-+A(u_++u_-))+2\nu_1=0$, which is (3.6). Let $v_\pm := (1\pm A)u_\pm$ be the Darcy fields of the two phases and $\delta := v_+ - v_-$. Then $v = u+Am = \lambda_+v_+ + \lambda_-v_-$, so $v_+ = v+\lambda_-\delta$, $v_- = v-\lambda_+\delta$, and

$$(1-A^2)m = (1-A)\lambda_+v_+ - (1+A)\lambda_-v_- = (\rho-A)v + 2\lambda_+\lambda_-\delta = (\rho-A)v + \tfrac12(1-\rho^2)\delta,$$

that is, $d=\delta$. The first condition in (3.6) reads $(1-A)v_+\cdot\nu=(1+A)v_-\cdot\nu$, i.e. $\delta\cdot\nu\,[(1-A)\lambda_-+(1+A)\lambda_+]=2A\,v\cdot\nu$; since $(1-A)\lambda_-+(1+A)\lambda_+=\mu$, this is $\delta\cdot\nu=\frac{2A}{\mu}v\cdot\nu$. The second condition reads $\delta\cdot\nu^\perp=-2\nu_1=-2e_2\cdot\nu^\perp$. Hence

$$\delta=\frac{2A}{\mu}(v\cdot\nu)\nu-2(e_2\cdot\nu^\perp)\nu^\perp=-2e_2+2\Big(\big(e_2+\tfrac{A}{\mu}v\big)\cdot\nu\Big)\nu,$$

where we used $\nu^\perp\otimes\nu^\perp=I-\nu\otimes\nu$. This is (3.7), and $|2(a\cdot\nu)\nu-a|^2=|a|^2$. The computation can be reversed: given (ρ,v,ν) put $\delta=-2e_2+2(a\cdot\nu)\nu$, $v_+=v+\lambda_-\delta$, $v_-=v-\lambda_+\delta$ and $u_\pm=v_\pm/(1\pm A)$. □

Proposition 3.2 (The relaxed set)**.** *The set*

$$\mathcal{U}_A:=\big\{(\rho,u,m)\in\mathbb{R}\times\mathbb{R}^2\times\mathbb{R}^2:\ |\rho|<1,\ |D-a|<|a|\big\}=\big\{|\rho|<1,\ D\cdot(D-2a)<0\big\},$$

with D,a given by (3.5), is open, and every point of $\mathcal{U}_A$ is the barycentre of a Λ-segment whose endpoints are first order laminates of K as in Lemma 3.1. In particular $\mathcal{U}_A$ is contained in the lamination convex hull K^{lc}, hence in the interior of the Λ-convex hull K^Λ.

Proof. Openness is clear. For $\bar m\neq0$ the state $\bar W=(0,-A\bar m,\bar m)$ belongs to Λ: take $\xi=\bar m^\perp$ and any c in (3.4). Moving along $\bar W$ leaves ρ and $v=u+Am$ (hence a) unchanged and moves d, hence D, in the direction $\bar m$. Let $(\rho,u,m)\in\mathcal{U}_A$; then D lies in the open disc $\{|D'-a|<|a|\}$, and any line through D meets the circle $\{|D'-a|=|a|\}$ at two points D_1,D_2 with $D\in(D_1,D_2)$. The circle is exactly $\{2(a\cdot\nu)\nu:\ |\nu|=1\}$ (the circle through the origin with centre a), so by the converse part of Lemma 3.1 the states with values (ρ,v,D_j) are first order laminates. The equivalence of the two descriptions of $\mathcal{U}_A$ follows from $|D-a|^2-|a|^2=D\cdot(D-2a)$. □

Remark 3.3. (a) For $A=0$ we have $v=u$, $a=e_2$ and $D-a=\frac{2}{1-\rho^2}\big(m-\rho u+\frac12(1-\rho^2)e_2\big)$, so $\mathcal{U}_0$ is the interior of the hull $\{|m-\rho u+\frac12(1-\rho^2)e_2|\le\frac12(1-\rho^2)\}$ computed by Székelyhidi [Sze12].

(b) The disc degenerates when $a=0$, i.e. $v=-\frac{\mu}{A}e_2$. For a state in a thin mixing zone around an interface, $a\cdot n$ is (up to lower order terms) $-\frac12\sigma$ and $a\cdot\tau$ is related to the vorticity of the sheet (see Lemma 5.2); thus mixing is impossible where there is neither Saffman–Taylor instability nor vorticity, in agreement with the pinch singularity observed in [Men22].

(c) We only need $\mathcal{U}_A\subset\operatorname{int}\mathsf{K}^\Lambda$; we do not claim that $\mathcal{U}_A$ coincides with the interior of the Λ-convex hull.

3.3. Subsolutions and the h-principle.

Definition 3.4. Let $T>0$ and $\rho_0\in L^\infty(\mathbb{T}\times\mathbb{R})$ with $\rho_0^2=1$. A triple $(\bar\rho,\bar u,\bar m)\in L^\infty(\mathbb{T}\times\mathbb{R}\times(0,T))^{1+2+2}$ is a *subsolution* with initial datum ρ_0 and mixing zone Ω_{mix}, an open subset of $\mathbb{T}\times\mathbb{R}\times(0,T)$, if

(i) (3.2) holds in the sense of distributions on $\mathbb{T}\times\mathbb{R}\times[0,T)$ with initial datum ρ_0, i.e. for every $\varphi\in C_c^\infty(\mathbb{T}\times\mathbb{R}\times[0,T))$

$$\iint\big(\bar\rho\partial_t\varphi+\bar m\cdot\nabla\varphi\big)+\int\rho_0\varphi(\cdot,0)=0,\qquad\iint\bar u\cdot\nabla\varphi=0,\qquad\iint\big((\bar u+A\bar m)\cdot\nabla^\perp\varphi+\bar\rho\,\partial_1\varphi\big)=0;$$

(ii) $(\bar\rho,\bar u,\bar m)$ is continuous on Ω_{mix} and takes values in $\mathcal{U}_A$ there;

(iii) $(\bar\rho,\bar u,\bar m)\in\mathsf{K}$ a.e. in the complement of Ω_{mix}.

The following h-principle was proved by Mengual [Men22] for $|A|<1$ (for $A=0$ see [CFG11, Sze12, CCF21]); we state it in the form in which we use it. The convex integration scheme is local in space-time, so the horizontally periodic setting plays no role, and by Proposition 3.2 subsolutions in the sense of Definition 3.4 take values in the interior of the Λ-convex hull, as required in [Men22]. We refer to [Men22] for the precise functional framework.

Theorem 3.5 (h-principle [Men22])**.** *Let $(\bar\rho,\bar u,\bar m)$ be a subsolution with initial datum ρ_0 and mixing zone Ω_{mix}. Then there exist infinitely many weak solutions (ρ,u) of (1.1) with initial datum ρ_0 such that $(\rho,u)=(\bar\rho,\bar u)$ a.e. outside Ω_{mix} and ρ mixes in Ω_{mix}, i.e. both $\{\rho=1\}$ and*

$\{\rho = -1\}$ have positive measure in every open subset of Ω_{mix}. Moreover there is a sequence of such solutions with $(\rho_k, u_k) \overset{*}{\rightharpoonup} (\bar\rho, \bar u)$ in L^∞.

3.4. **The flat interface.** The following explicit subsolutions illustrate the role of the viscosity jump; the flat case is treated in [Men22].

Example 3.6. Let $\theta \in (0,1)$ and $F_\theta(\rho) := \theta\frac{1-\rho^2}{1-A\rho}$ for $\rho \in [-1,1]$. Then $F_\theta(\pm1) = 0$,

$$F_\theta'(\rho) = \theta\,\frac{A - 2\rho + A\rho^2}{(1-A\rho)^2}, \qquad F_\theta''(\rho) = -\frac{2\theta(1-A^2)}{(1-A\rho)^3} < 0,$$

so $-F_\theta'$ increases from $-\frac{2\theta}{1+A}$ at $\rho=-1$ to $\frac{2\theta}{1-A}$ at $\rho=1$. Let $R_\theta : \mathbb{R} \to [-1,1]$ be equal to -1 on $(-\infty, -\frac{2\theta}{1+A}]$, to 1 on $[\frac{2\theta}{1-A}, \infty)$, and to the inverse of $-F_\theta'$ in between, and set

$$\bar\rho(x,t) = R_\theta(x_2/t), \qquad \bar u = 0, \qquad \bar m = -F_\theta(\bar\rho)\,e_2.$$

Then $\partial_t\bar\rho + \partial_2\bar m_2 = -\frac{R_\theta'}{t}\big(\frac{x_2}{t} + F_\theta'(R_\theta)\big) = 0$ (a rarefaction wave), $\operatorname{div}\bar u = 0$, and $\operatorname{curl}(\bar u + A\bar m) = -A\partial_1 F_\theta(\bar\rho) = 0 = -\partial_1\bar\rho$; all fields are continuous, so (3.2) holds with initial datum $\operatorname{sign}(x_2)$. Outside the fan $\{-\frac{2\theta t}{1+A} < x_2 < \frac{2\theta t}{1-A}\}$ the state is $(\pm1, 0, 0) \in \mathsf{K}$. Inside, $v = A\bar m$, $(1-A^2)\bar m - (\bar\rho - A)v = (1-A\bar\rho)\bar m$, hence

$$D = 2(1-\theta)e_2, \qquad a = \Big(1 - \frac{A^2\theta(1-\bar\rho^2)}{1-A^2\bar\rho^2}\Big)e_2, \qquad D\cdot(D-2a) = -\frac{4\theta(1-\theta)(1-A^2)}{1-A^2\bar\rho^2} < 0.$$

So $(\bar\rho, \bar u, \bar m)$ is a subsolution, and Theorem 3.5 gives mixing solutions whose mixing zone is $\{-\frac{2\theta t}{1+A} < x_2 < \frac{2\theta t}{1-A}\}$. For $A > 0$ (the heavy fluid on top is the more viscous one) the mixing zone grows faster upwards: the less viscous light fluid fingers into the more viscous one, as predicted by Saffman and Taylor. Moreover $\int(\bar\rho - \operatorname{sign} x_2)\,\mathrm{d}x_2 = t\int_{-1}^{1} F_\theta'(r)\,\mathrm{d}r = 0$: the centre of mass of the mixing zone does not move.

4. Piecewise constant subsolutions

In this section $z_0 \in \mathcal{Z}^{6,\gamma}$, $\rho_{\mathrm m} \in (-1,1)$ and $\kappa \in C^{4,\gamma}(\mathbb{T})$ with $\kappa > 0$ are fixed, $\sigma_0, U_0 \in C^{4,\gamma}(\mathbb{T})$ are as in Definition 2.10, and

$$\mu_{\mathrm m} := 1 + A\rho_{\mathrm m}, \qquad c^\pm := (1 \pm \rho_{\mathrm m})\kappa.$$

Moreover $a \in C^{3,\gamma}(\mathbb{T})$ and a polynomial h with $h(0) = h'(0) = 0$ are given; they will be chosen in Section 5.

4.1. **Geometry.** We define the central curve and the two boundary curves of the mixing zone,

(4.1)
$$Z(\alpha,t) := z_0(\alpha) + \Big(tU_0(\alpha) + \tfrac{t^2}{2}a(\alpha)\Big)n_0(\alpha) + h(t)e_2, \qquad z^\pm(\alpha,t) := Z(\alpha,t) \pm t\,c^\pm(\alpha)\,n_0(\alpha),$$

and the map

(4.2)
$$X(\alpha,\lambda,t) := \tfrac12(1-\lambda)\,z^-(\alpha,t) + \tfrac12(1+\lambda)\,z^+(\alpha,t), \qquad \partial_\lambda X = \tfrac12(z^+ - z^-) = t\kappa\,n_0.$$

Lemma 4.1. *There is $T_1 > 0$ such that for $t \in (0,T_1]$: $z^\pm(\cdot,t) \in \mathcal{Z}^{3,\gamma}$ with $\delta(z^\pm(\cdot,t)) \ge \frac12\delta(z_0)$; $\Gamma^+(t) \subset \Omega^+(z^-(\cdot,t))$; and $X(\cdot,\cdot,t)$ is a $C^{3,\gamma}$ diffeomorphism from $\mathbb{T}\times[-1,1]$ onto the closure of the strip*

$$S_t := \Omega^-(z^+(\cdot,t)) \cap \Omega^+(z^-(\cdot,t)),$$

mapping $\lambda = \pm1$ onto $\Gamma^\pm(t)$. Moreover $z^\pm$ are polynomials in t with coefficients in $C^{3,\gamma}$, and $z^\pm(\cdot,t) \to z_0$ in $C^{3,\gamma}$ as $t \to 0$.

Proof. All statements follow from $z^\pm(\cdot,t) - z_0 = O(t)$ in $C^{3,\gamma}$, the continuity of δ with respect to the C^1 norm, $z^+ - z^- = 2t\kappa n_0$ with $\kappa > 0$, and the fact that $\partial_\alpha X = \partial_\alpha z_0 + O(t)$ and $\partial_\lambda X = t\kappa n_0$ are linearly independent; injectivity of X for small t is the usual tubular neighbourhood argument. □

We denote by $n^\pm$ the upward unit normals of $\Gamma^\pm(t)$ and by

$$V^\pm := \partial_t z^\pm \cdot n^\pm$$

their normal velocities. Note that the central curve Z is the *centre of mass* of the strip: since $(1-\rho_{\mathrm m})c^+ = (1+\rho_{\mathrm m})c^-$,

$$Z = \tfrac12\big[(1-\rho_{\mathrm m})z^+ + (1+\rho_{\mathrm m})z^-\big]. \tag{4.3}$$

4.2. **The subsolution.** Let $[\![\rho]\!]^+ := 1-\rho_{\mathrm m}$ and $[\![\rho]\!]^- := 1+\rho_{\mathrm m}$ be the jumps of the density across $\Gamma^\pm$, and define

$$f^\pm := [\![\rho]\!]^\pm\Big(-e_2 + \big(n_2^\pm + AV^\pm\big)n^\pm\Big), \qquad v := \mathcal C[z^+;f^+] + \mathcal C[z^-;f^-] \quad \text{on } \mathbb R^2\setminus(\Gamma^+\cup\Gamma^-). \tag{4.4}$$

Let $v_S^\pm(\alpha,t)$ be the limits of v at $z^\pm(\alpha,t)$ from inside the strip:

$$v_S^+ = \mathcal C[z^+;f^+]^- + \mathcal C[z^-;f^-]\circ z^+, \qquad v_S^- = \mathcal C[z^-;f^-]^+ + \mathcal C[z^+;f^+]\circ z^-. \tag{4.5}$$

We define the *boundary fluxes* and the *flux mismatch*

$$F^+ := |\partial_\alpha z^+|\Big\{\tfrac{2}{1+\rho_{\mathrm m}}\big(v_S^+\cdot n^+ - (1-A)V^+\big) + 2n_2^+\Big\}, \tag{4.6}$$

$$F^- := |\partial_\alpha z^-|\Big\{\tfrac{2}{1-\rho_{\mathrm m}}\big((1+A)V^- - v_S^-\cdot n^-\big) + 2n_2^-\Big\}, \tag{4.7}$$

$$M := F^+ - F^-, \tag{4.8}$$

the functions $\Psi^\pm(\alpha,t) := \int_0^\alpha F^\pm(\beta,t)\,\mathrm d\beta$, and the stream function Ψ on $\overline{S_t}$ by

$$\Psi\big(X(\alpha,\lambda,t),t\big) := \tfrac12(1-\lambda)\Psi^-(\alpha,t) + \tfrac12(1+\lambda)\Psi^+(\alpha,t). \tag{4.9}$$

Finally we set $\bar\rho(\cdot,0)=\rho_0$ and, for $0<t\le T_1$,

$$(\bar\rho,\bar u,\bar m) := \begin{cases} \Big(1,\ \dfrac{v}{1+A},\ \dfrac{v}{1+A}\Big) & \text{in } \Omega^+(z^+),\\[2mm] \Big(\rho_{\mathrm m},\ v - A\bar m,\ \dfrac{(\rho_{\mathrm m}-A)v + \frac12(1-\rho_{\mathrm m}^2)(D-2e_2)}{1-A^2}\Big), \quad D := \nabla^\perp\Psi, & \text{in } S_t,\\[2mm] \Big(-1,\ \dfrac{v}{1-A},\ -\dfrac{v}{1-A}\Big) & \text{in } \Omega^-(z^-). \end{cases} \tag{4.10}$$

By construction, in S_t the Darcy field of the state is $\bar u + A\bar m = v$, and the quantities (3.5) are $d = D - 2e_2$, $D = \nabla^\perp\Psi$ and $a = e_2 + \frac{A}{\mu_{\mathrm m}}v$.

Lemma 4.2 (The linear equations)**.** *Assume that*

$$\int_{\mathbb T} M(\alpha,t)\,\mathrm d\alpha = 0 \qquad \textit{for all } t\in(0,T_1]. \tag{4.11}$$

Then D is well defined and 2π-periodic in x_1, $(\bar\rho,\bar u,\bar m)$ is bounded, and it solves (3.2) in the sense of distributions on $\mathbb T\times\mathbb R\times[0,T_1)$ with initial datum ρ_0. Outside the closure of $\bigcup_t S_t\times\{t\}$ it takes values in $\mathsf K$.

Proof. Periodicity. $\Psi^\pm(\alpha+2\pi,t) = \Psi^\pm(\alpha,t) + \Phi^\pm(t)$ with $\Phi^\pm = \int_{\mathbb T} F^\pm$, and $\Phi^+ = \Phi^-$ by (4.11). Since $X(\alpha+2\pi,\lambda,t) = X(\alpha,\lambda,t) + 2\pi e_1$, we get $\Psi(x+2\pi e_1) = \Psi(x) + \Phi^+$, so $D = \nabla^\perp\Psi$ is periodic. Ψ is C^1 up to ∂S_t, hence D is bounded.

The equations in the open regions. In each of the three open regions $\bar\rho$ is constant; v is divergence-free and curl-free there by Lemma 2.3(a), and $\operatorname{div} D = \operatorname{div}\nabla^\perp\Psi = 0$. Hence $\operatorname{div}\bar m = 0$, $\operatorname{div}\bar u = \operatorname{div} v - A\operatorname{div}\bar m = 0$ and $\operatorname{curl}(\bar u + A\bar m) = \operatorname{curl} v = 0$ in each region. Outside the strip, $\bar m = \bar\rho\bar u$ with $\bar\rho = \pm1$, i.e. the state is in $\mathsf K$.

Jump conditions. A piecewise C^1 bounded field satisfies (3.2) in $\mathcal D'$ if and only if it does so in each open region and satisfies the Rankine–Hugoniot conditions across the moving curves:

$$[\![\bar\rho]\!]V = [\![\bar m\cdot n]\!], \qquad [\![\bar u\cdot n]\!] = 0, \qquad [\![(\bar u + A\bar m)\cdot\tau]\!] = -[\![\bar\rho]\!]\tau_2.$$

(The initial condition holds because the fields are bounded and $\bar\rho(\cdot,t)\to\rho_0$ in L^1_{loc}.) Consider Γ^+. The jump of v across Γ^+ is f^+ (the field $\mathcal C[z^-;f^-]$ is continuous there), and $f^+\cdot\tau^+ =$

$-[\![\rho]\!]^+\tau_2^+$, $f^+\cdot n^+ = A[\![\rho]\!]^+V^+$. Since $\bar u + A\bar m = v$ on both sides, the third condition holds. Next, the flux of $\nabla^\perp\Psi$ through Γ^+ per unit α is $\nabla\Psi\cdot\partial_\alpha z^+ = \partial_\alpha(\Psi\circ z^+) = \partial_\alpha\Psi^+ = F^+$, so on the strip side

$$D\cdot n^+ = \frac{F^+}{|\partial_\alpha z^+|}, \qquad (D-2e_2)\cdot n^+ = \frac{2}{1+\rho_{\mathrm m}}\big(v_S^+\cdot n^+ - (1-A)V^+\big),$$

and therefore

$$(1-A^2)\,\bar m^S\cdot n^+ = (\rho_{\mathrm m}-A)v_S^+\cdot n^+ + (1-\rho_{\mathrm m})\big(v_S^+\cdot n^+ - (1-A)V^+\big) = (1-A)\big(v_S^+\cdot n^+ - (1-\rho_{\mathrm m})V^+\big).$$

Above Γ^+, $\bar m\cdot n^+ = v^{\mathrm{above}}\cdot n^+/(1+A) = (v_S^+\cdot n^+ + A(1-\rho_{\mathrm m})V^+)/(1+A)$. Subtracting, $[\![\bar m\cdot n^+]\!] = (1-\rho_{\mathrm m})V^+ = [\![\bar\rho]\!]V^+$. Moreover $\bar u^S\cdot n^+ = v_S^+\cdot n^+ - A\bar m^S\cdot n^+ = (v_S^+\cdot n^+ + A(1-\rho_{\mathrm m})V^+)/(1+A) = \bar u^{\mathrm{above}}\cdot n^+$. The computation on Γ^- is identical: there $(D-2e_2)\cdot n^- = \frac{2}{1-\rho_{\mathrm m}}((1+A)V^- - v_S^-\cdot n^-)$, which gives $(1-A^2)\bar m^S\cdot n^- = (1+A)\big((1+\rho_{\mathrm m})V^- - v_S^-\cdot n^-\big)$, while below Γ^-, $\bar m\cdot n^- = -(v_S^-\cdot n^- - A(1+\rho_{\mathrm m})V^-)/(1-A)$; hence $[\![\bar m\cdot n^-]\!] = (1+\rho_{\mathrm m})V^-$ and $[\![\bar u\cdot n^-]\!] = 0$. □

Thus the ansatz (4.10) solves the linear system exactly, whatever the curves $z^\pm$ are, provided (4.11) holds; the only remaining requirement for a subsolution is the open condition $D\cdot(D-2a)<0$ in S_t. The next lemma identifies (4.11) with the conservation of mass.

Lemma 4.3 (Mass identity)**.** *Let* $\mathcal{A}^\pm(t) := \int_{\mathbb T} z_2^\pm\,\partial_\alpha z_1^\pm\,\mathrm d\alpha$, *so that* $\frac{\mathrm d}{\mathrm dt}\mathcal{A}^\pm = \int_{\mathbb T} V^\pm|\partial_\alpha z^\pm|\,\mathrm d\alpha$, *and*

$$Q(t) := (1-\rho_{\mathrm m})\tfrac{\mathrm d}{\mathrm dt}\mathcal{A}^+(t) + (1+\rho_{\mathrm m})\tfrac{\mathrm d}{\mathrm dt}\mathcal{A}^-(t).$$

Then for $t\in(0,T_1]$

$$\int_{\mathbb T} M(\alpha,t)\,\mathrm d\alpha = -\frac{2\,Q(t)}{1-\rho_{\mathrm m}^2}.$$

Proof. The identity for $\frac{\mathrm d}{\mathrm dt}\mathcal{A}^\pm$ follows from an integration by parts: $\frac{\mathrm d}{\mathrm dt}\mathcal{A} = \int(\partial_t z_2\partial_\alpha z_1 - \partial_\alpha z_2\partial_t z_1) = \int\partial_t z\cdot(\partial_\alpha z)^\perp$. By Lemma 2.3(c), the constant parts $-[\![\rho]\!]^\pm e_2$ of the densities contribute $\mp\frac12\sum_\pm[\![\rho]\!]^\pm e_2 = \mp e_2$ to $v_{\pm\infty}$, while by (2.2) the normal parts contribute $\pm\frac{1}{4\pi}\sum_\pm[\![\rho]\!]^\pm(2\pi + A\frac{\mathrm d}{\mathrm dt}\mathcal{A}^\pm)e_2$. Hence

$$v_{\pm\infty} = \pm\frac{AQ}{4\pi}e_2.$$

Since v is divergence-free and periodic above Γ^+, the flux of v through Γ^+ from above equals $2\pi(v_{+\infty})_2 = \frac{AQ}{2}$; subtracting the jump $f^+\cdot n^+ = A(1-\rho_{\mathrm m})V^+$,

$$\int_{\mathbb T} v_S^+\cdot n^+|\partial_\alpha z^+|\,\mathrm d\alpha = \frac{AQ}{2} - A(1-\rho_{\mathrm m})\tfrac{\mathrm d}{\mathrm dt}\mathcal{A}^+, \qquad \int_{\mathbb T} v_S^-\cdot n^-|\partial_\alpha z^-|\,\mathrm d\alpha = -\frac{AQ}{2} + A(1+\rho_{\mathrm m})\tfrac{\mathrm d}{\mathrm dt}\mathcal{A}^-,$$

the second identity being obtained in the same way from below. Using $\int n_2^\pm|\partial_\alpha z^\pm| = 2\pi$ we get

$$\int_{\mathbb T} F^+ = \frac{2}{1+\rho_{\mathrm m}}\Big(\frac{AQ}{2} - (1-A\rho_{\mathrm m})\tfrac{\mathrm d}{\mathrm dt}\mathcal{A}^+\Big) + 4\pi,$$
$$\int_{\mathbb T} F^- = \frac{2}{1-\rho_{\mathrm m}}\Big((1-A\rho_{\mathrm m})\tfrac{\mathrm d}{\mathrm dt}\mathcal{A}^- + \frac{AQ}{2}\Big) + 4\pi,$$

and the difference is $\frac{2}{1-\rho_{\mathrm m}^2}\big(-\rho_{\mathrm m}AQ - (1-A\rho_{\mathrm m})Q\big) = -\frac{2Q}{1-\rho_{\mathrm m}^2}$. □

Remark 4.4. $Q=0$ means that $(1-\rho_{\mathrm m})\mathcal{A}^+ + (1+\rho_{\mathrm m})\mathcal{A}^-$ is constant, i.e. that the total mass $\int(\bar\rho-\rho_{\mathrm{ref}})$ over a period is conserved. In view of (4.3), the central interface Z separates, to first order in the width of the strip, the same amounts of heavy and light fluid as the strip itself; this is why Z must move, to leading order, with the normal velocity of the Muskat problem (Lemma 5.1).

The following lemma shows that the tangential component of D is controlled by the primitive of M, and its normal component by the boundary fluxes.

Lemma 4.5 (The state in the strip)**.** *Assume* (4.11)*, and that there are* $C_1 \geq 1$ *and* $F_0 \in C^0(\mathbb{T})$ *such that for* $t \in (0, T_1]$

$$\|M(\cdot,t)\|_{C^0(\mathbb{T})} \leq C_1 t^2, \qquad \|F^\pm(\cdot,t) - F_0\|_{C^0(\mathbb{T})} \leq C_1 t.$$

Then there is C *such that, for* $t \in (0, T_1]$ *and* $x = X(\alpha,\lambda,t) \in S_t$,

$$\Big| D(x,t) - \frac{F_0(\alpha)}{|\partial_\alpha z_0(\alpha)|}\, n_0(\alpha) \Big| \leq Ct.$$

Proof. Differentiating (4.9) and using (4.2),

$$t\kappa\, \nabla\Psi \cdot n_0 = \partial_\lambda\big(\Psi \circ X\big) = \tfrac12(\Psi^+ - \Psi^-) = \tfrac12 \int_0^\alpha M(\beta,t)\,\mathrm{d}\beta,$$
$$\nabla\Psi \cdot \partial_\alpha X = \partial_\alpha\big(\Psi \circ X\big) = \tfrac12(1-\lambda)F^- + \tfrac12(1+\lambda)F^+.$$

Hence $|\nabla\Psi \cdot n_0| \leq \pi C_1 t/\min\kappa$, and since $\partial_\alpha X = |\partial_\alpha z_0|\tau_0 + O(t)$, $\nabla\Psi\cdot\tau_0 = F_0/|\partial_\alpha z_0| + O(t)$. As $D = \nabla^\perp\Psi = (\nabla\Psi\cdot\tau_0)n_0 - (\nabla\Psi\cdot n_0)\tau_0$, the claim follows. □

5. Proof of Theorem 1.1

We keep the notation of Section 4. The proof consists in choosing the normal acceleration a and the vertical correction h in (4.1) so that $M = O(t^2)$ and (4.11) holds, and then in checking the open condition $D\cdot(D-2a) < 0$ in the strip by comparison with its limit as $t \to 0$.

5.1. **The limit $t \to 0$: velocity matching.** For the computation of the limits it is convenient to allow arbitrary normal velocities at $t = 0$.

Lemma 5.1. *Let* $z^\pm(\cdot,t)$ *be polynomials in* t *with coefficients in* $C^{3,\gamma}$*, with* $z^\pm(\cdot,0) = z_0$ *and* $\partial_t z^\pm(\cdot,0)\cdot n_0 = W^\pm \in C^{3,\gamma}(\mathbb{T})$*, such that the conclusions of Lemma 4.1 hold, and define* $f^\pm, v, F^\pm, M$ *by* (4.4)–(4.8)*. Let*

$$\overline{W} := \tfrac12\big[(1-\rho_{\mathrm m})W^+ + (1+\rho_{\mathrm m})W^-\big], \qquad P := 2\mathcal{K}_0[n_{0,2}] + 2A\mathcal{K}_0\overline{W}.$$

Then, uniformly on $\mathbb{T}$ *as* $t \to 0$, $v_S^\pm(\cdot,t) \to v_S^{(0)} := \mathcal{C}[z_0; f_0^+]^- + \mathcal{C}[z_0; f_0^-]^+$*, where* $f_0^\pm = [\![\rho]\!]^\pm(-e_2 + (n_{0,2} + AW^\pm)n_0)$*, and*

$$v_S^{(0)}\cdot n_0 = P + \tfrac{A}{2}\big[(1+\rho_{\mathrm m})W^- - (1-\rho_{\mathrm m})W^+\big], \tag{5.1}$$

$$M(\cdot,t) \to M_0 := -\frac{4|\partial_\alpha z_0|}{1-\rho_{\mathrm m}^2}\Big[\big(I - 2A\mathcal{K}_0\big)\overline{W} - 2\mathcal{K}_0[n_{0,2}]\Big]. \tag{5.2}$$

For the ansatz (4.1)*, i.e.* $W^\pm = U_0 \pm c^\pm$*, we have* $\overline{W} = U_0$*,* $P = U_0$*,* $M_0 = 0$*,*

$$v_S^{(0)}\cdot n_0 = \mu_{\mathrm m}U_0 - A(1-\rho_{\mathrm m}^2)\kappa, \qquad F^\pm(\cdot,t) \to F_0 := |\partial_\alpha z_0|\big(-\sigma_0 - 2(1-A\rho_{\mathrm m})\kappa\big). \tag{5.3}$$

Proof. The convergence of $v_S^\pm$ follows from Lemma A.2: for instance $\mathcal{C}[z^+; f^+]^- \to \mathcal{C}[z_0; f_0^+]^-$, and $\mathcal{C}[z^-; f^-](z^+(\alpha,t)) \to \mathcal{C}[z_0; f_0^-]^+(\alpha)$ because $z^+(\alpha,t)$ lies above $\Gamma^-(t)$ and tends to $z_0(\alpha)$. By Lemma 2.3(b), $v_S^{(0)} = \langle \mathcal{C}[z_0; f_0^+ + f_0^-]\rangle - \frac12 f_0^+ + \frac12 f_0^-$, and $f_0^+ + f_0^- = -2e_2 + 2(n_{0,2} + A\overline{W})n_0$ because $[\![\rho]\!]^+ + [\![\rho]\!]^- = 2$. Since $\langle \mathcal{C}[z_0; e_2]\rangle = 0$ by Lemma 2.3(c), $\langle \mathcal{C}[z_0; f_0^+ + f_0^-]\rangle \cdot n_0 = P$, and (5.1) follows from $f_0^\pm \cdot n_0 = A[\![\rho]\!]^\pm W^\pm$. Passing to the limit in (4.6)–(4.7),

$$\frac{M_0}{|\partial_\alpha z_0|} = \frac{2}{1-\rho_{\mathrm m}^2}\Big\{2v_S^{(0)}\cdot n_0 - (1-\rho_{\mathrm m})(1-A)W^+ - (1+\rho_{\mathrm m})(1+A)W^-\Big\} = \frac{4}{1-\rho_{\mathrm m}^2}\big(P - \overline{W}\big),$$

which is (5.2). For the ansatz, $(1-\rho_{\mathrm m})c^+ = (1+\rho_{\mathrm m})c^- = (1-\rho_{\mathrm m}^2)\kappa$ gives $\overline{W} = U_0$. Applying $\mathcal{K}_0$ to $(I - 2A\mathcal{K}_0)\sigma_0 = -2n_{0,2}$ and using $U_0 = -\mathcal{K}_0\sigma_0$ we get $(I - 2A\mathcal{K}_0)U_0 = 2\mathcal{K}_0[n_{0,2}]$, i.e. $P = U_0$ and $M_0 = 0$. Then (5.1) gives $v_S^{(0)}\cdot n_0 = U_0 + A\rho_{\mathrm m}U_0 - A(1-\rho_{\mathrm m}^2)\kappa$, and inserting this in (4.6) and using $\sigma_0 = -2(n_{0,2} + AU_0)$ (Proposition 2.7),

$$\frac{F_0}{|\partial_\alpha z_0|} = \frac{2}{1+\rho_{\mathrm m}}\Big(A(1+\rho_{\mathrm m})U_0 - (1+\rho_{\mathrm m})\kappa\big[A(1-\rho_{\mathrm m}) + 1 - A\big]\Big) + 2n_{0,2} = -\sigma_0 - 2(1-A\rho_{\mathrm m})\kappa.$$ □

Thus the vanishing of M at order t^0 is precisely the statement that the centre of mass Z of the strip starts with the normal velocity of the Muskat problem, which we built into the ansatz.

5.2. **The relaxation condition at $t = 0$.**

Lemma 5.2. *Let $D^{(0)} := \frac{F_0}{|\partial_\alpha z_0|} n_0 = \big(-\sigma_0 - 2(1 - A\rho_{\mathrm{m}})\kappa\big) n_0$ and $a^{(0)} := e_2 + \frac{A}{\mu_{\mathrm{m}}} v_S^{(0)}$. Then*

$$a^{(0)} \cdot n_0 = -\frac{\sigma_0}{2} - \frac{A^2(1-\rho_{\mathrm{m}}^2)\kappa}{\mu_{\mathrm{m}}}, \qquad D^{(0)} \cdot \big(D^{(0)} - 2a^{(0)}\big) = -\frac{2\kappa(1-A^2)}{\mu_{\mathrm{m}}}\Big(-\sigma_0 - 2(1-A\rho_{\mathrm{m}})\kappa\Big).$$

In particular, under (1.7)–(1.8), $\max_{\mathbb{T}} D^{(0)} \cdot (D^{(0)} - 2a^{(0)}) =: -m_0 < 0$.

Proof. By (5.3), $a^{(0)} \cdot n_0 = n_{0,2} + AU_0 - \frac{A^2(1-\rho_{\mathrm{m}}^2)\kappa}{\mu_{\mathrm{m}}}$, and $n_{0,2} + AU_0 = -\sigma_0/2$. Since $D^{(0)}$ is normal, $D^{(0)} \cdot (D^{(0)} - 2a^{(0)}) = D_n(D_n - 2a_n)$ with $D_n = -\sigma_0 - 2(1 - A\rho_{\mathrm{m}})\kappa$ and $a_n = a^{(0)} \cdot n_0$, and

$$D_n - 2a_n = -2\kappa\Big[(1 - A\rho_{\mathrm{m}}) - \frac{A^2(1-\rho_{\mathrm{m}}^2)}{1 + A\rho_{\mathrm{m}}}\Big] = -\frac{2\kappa(1-A^2)}{\mu_{\mathrm{m}}}. \qquad \square$$

Remark 5.3. Lemma 5.2 explains the range (1.8): the normal component of the relaxed flux, D_n, must lie strictly between 0 and $2a_n$; the upper bound holds automatically because $|A| < 1$, and the lower bound is $2(1 - A\rho_{\mathrm{m}})\kappa < -\sigma_0$. When $\sigma_0 \geq 0$ somewhere, no $\kappa > 0$ is admissible there; this is the Saffman–Taylor stability of the interface.

5.3. **Time regularity and acceleration matching.** The following lemma is proved in Appendix A.

Lemma 5.4. *Let $a \in C^{3,\gamma}(\mathbb{T})$ and let h be a polynomial with $h(0) = h'(0) = 0$. There are $T_1 > 0$ and C such that $F^\pm$ and M, extended to $t = 0$ by their limits, belong to $C^1([0, T_1]; C^0(\mathbb{T}))$, $\|\partial_t F^\pm(\cdot, t) - \partial_t F^\pm(\cdot, s)\|_{C^0} \leq C|t - s|$, and*

$$\sup_{x = X(\alpha,\lambda,t) \in S_t} \big|v(x,t) - v_S^{(0)}(\alpha)\big| \leq Ct^\gamma \qquad (0 < t \leq T_1). \tag{5.4}$$

Lemma 5.5. *There is $G \in C^{3,\gamma}(\mathbb{T})$, depending only on $z_0, \kappa, \rho_{\mathrm{m}}$ and A, such that for every $a \in C^{3,\gamma}(\mathbb{T})$ and $h \equiv 0$,*

$$\partial_t M(\cdot, 0) = G - \frac{4|\partial_\alpha z_0|}{1 - \rho_{\mathrm{m}}^2}\big(I - 2A\mathcal{K}_0\big)a.$$

Proof. For fixed t the quantities $V^\pm$, $f^\pm$, v, $v_S^\pm$, $F^\pm$ and M are affine functions of the velocities $(\partial_t z^+, \partial_t z^-)$, with coefficients depending only on the positions $z^\pm(\cdot, t)$: indeed $V^\pm$ is linear in $\partial_t z^\pm$, $f^\pm$ is affine in $V^\pm$, and v is linear in $f^\pm$. Write accordingly $M(t) = \mathcal{L}_t[\partial_t z^+, \partial_t z^-] + \mathcal{B}_t$. By the formulas of Appendix A, $\mathcal{L}_t$ and $\mathcal{B}_t$ are differentiable at $t = 0$ and their derivatives only involve z_0 and $\partial_t z^\pm(\cdot, 0) = (U_0 \pm c^\pm)n_0$, which do not depend on a (the term $\frac{t^2}{2} a n_0$ in the positions does not contribute at $t = 0$). Hence $\partial_t M(\cdot, 0) = \mathcal{L}_0[\partial_t^2 z^+, \partial_t^2 z^-](\cdot, 0) + G$ with G independent of a. Since $\partial_t^2 z^\pm(\cdot, 0) = a n_0$ and, by (5.2), the linear part of the map $(W^+, W^-) \mapsto M_0$ is $-\frac{4|\partial_\alpha z_0|}{1-\rho_{\mathrm{m}}^2}(I - 2A\mathcal{K}_0)\overline{W}$, we obtain the formula. The regularity $G \in C^{3,\gamma}$ is proved in Appendix A. $\square$

This is the second place where the operator $I - 2A\mathcal{K}(z_0)$ appears: the vanishing of M at order t^1 is an equation for the normal acceleration of the centre of mass, which we solve by Proposition 2.6. Note that only the restriction of $\mathcal{K}$ to z_0 is needed; the instability of the Muskat problem plays no role in a time expansion of finite order.

Lemma 5.6 (Mass correction)**.** *Let $a \in C^{3,\gamma}$ be such that $\partial_t M(\cdot, 0) = 0$ when $h \equiv 0$, and let $Q^{(0)}$ be the function Q of Lemma 4.3 computed with $h \equiv 0$. Then $h(t) := -\frac{1}{4\pi}\int_0^t Q^{(0)}(s)\,\mathrm{d}s$ is a polynomial with $|h(t)| \leq Ct^3$, and the curves (4.1) with this h satisfy (4.11) and $\|M(\cdot, t)\|_{C^0} \leq Ct^2$ for $t \in (0, T_1]$.*

Proof. Let $M^{(0)}$ be the mismatch for $h \equiv 0$. By Lemmas 5.1 and 5.4, $M^{(0)}(\cdot,0) = 0$, $\partial_t M^{(0)}(\cdot,0) = 0$ and $\partial_t M^{(0)}$ is Lipschitz in time, so $\|M^{(0)}(\cdot,t)\|_{C^0} \le Ct^2$, and by Lemma 4.3 $|Q^{(0)}(t)| \le Ct^2$. The areas $\mathcal{A}^\pm$ are polynomials in t (the curves are), hence so are $Q^{(0)}$ and h, and $h(0) = h'(0) = 0$, $|h'(t)| \le Ct^2$, $|h(t)| \le Ct^3$. Adding $h(t)e_2$ to both curves adds $2\pi h$ to $\mathcal{A}^\pm$, so $Q = Q^{(0)} + 4\pi h' = 0$ and (4.11) holds by Lemma 4.3. Finally, Cauchy fields commute with translations, $\mathcal{C}[z + he_2; f](x + he_2) = \mathcal{C}[z; f](x)$, so the new $F^\pm$ are obtained from the old ones by adding $h'e_2$ to the velocities; by the affine dependence on the velocities and Lemma 2.3(b), $\|M - M^{(0)}\|_{C^0} \le C|h'(t)| \le Ct^2$. □

5.4. **Conclusion.**

Proof of Theorem 1.1. Let G be as in Lemma 5.5 and define

$$a := \big(I - 2A\mathcal{K}_0\big)^{-1}\Big[\frac{(1-\rho_{\mathrm{m}}^2)\,G}{4|\partial_\alpha z_0|}\Big] \in C^{3,\gamma}(\mathbb{T}),$$

which is possible by Proposition 2.6 with $k = 6$, $j = 3$. Then $\partial_t M(\cdot,0) = 0$ when $h \equiv 0$. Let h be given by Lemma 5.6 and consider the curves (4.1) and the triple $(\bar\rho, \bar u, \bar m)$ of (4.10). By Lemma 5.6, (4.11) holds and $\|M\|_{C^0} \le Ct^2$; by Lemma 5.4 and since $h'(0) = 0$, $\|F^\pm(\cdot,t) - F_0\|_{C^0} \le Ct$ with F_0 given by (5.3). By Lemma 4.2, $(\bar\rho, \bar u, \bar m)$ is bounded, solves (3.2) with initial datum ρ_0, and lies in K outside the strip. By Lemma 4.5, $|D - D^{(0)}(\alpha)| \le Ct$ in S_t, and by (5.4), $|a - a^{(0)}(\alpha)| \le \frac{|A|}{\mu_{\mathrm{m}}} Ct^\gamma$ in S_t, where $a = e_2 + \frac{A}{\mu_{\mathrm{m}}} v$ is the vector (3.5) of the state in the strip. Since all these quantities are bounded, Lemma 5.2 gives in S_t

$$D \cdot (D - 2a) \le D^{(0)} \cdot \big(D^{(0)} - 2a^{(0)}\big) + C't^\gamma \le -m_0 + C't^\gamma < 0$$

for $0 < t \le T$, provided $T \le T_1$ and $C'T^\gamma < m_0$. Hence, with $\Omega_{\mathrm{mix}} := \{(x,t) : 0 < t < T,\ x \in S_t\}$, which is open, $(\bar\rho, \bar u, \bar m)$ is continuous on Ω_{mix} with values in $\mathcal{U}_A$ (Proposition 3.2), i.e. a subsolution in the sense of Definition 3.4. Theorem 3.5 yields infinitely many weak solutions with $\rho = \bar\rho$ outside Ω_{mix}, mixing in Ω_{mix}, and a sequence converging weakly-$*$ to $\bar\rho$. Since $\Omega_{\mathrm{mix}}(t) = S_t$ is the strip between $\Gamma^\pm(t)$, and Z has the form (1.9), all the assertions of Theorem 1.1 follow. □

6. Remarks and open problems

6.1. **Consistency with the case $A = 0$.** For $A = 0$ we have $\sigma_0 = -2n_{0,2}$ and (1.8) reads $0 < \kappa < n_{0,2}$. The strip has normal width $2t\kappa$, i.e. vertical width $2t\kappa/n_{0,2} < 2t$ (the curves are graphs). This is consistent with the maximal vertical opening speed 2 of the mixing zone for equal viscosities in our normalisation [Ott99, Sze12, CCF21, FS18].

6.2. **Optimal opening rates (formal).** Single-strip subsolutions do not exhaust the relaxation. Consider instead a mixing zone of width $O(t)$ around Z in which the density has a continuous profile $\bar\rho = R(\xi)$, $\xi = (x - Z)\cdot n/t$, R increasing from -1 at ξ_- to 1 at ξ_+, normalised so that Z is the centre of mass, $\int_{\xi_-}^{\xi_+} \xi R'(\xi)\,\mathrm{d}\xi = 0$. Repeating the computations of Section 5 at leading order (the conservation law now determines the normal component of D by an integration across the zone instead of the boundary fluxes (4.6)–(4.7)), one finds, with $\mathcal{P}(\xi) := -\int_{\xi_-}^{\xi} \eta R'(\eta)\,\mathrm{d}\eta \ge 0$ and $D \cdot \tau = 0$,

$$D \cdot n = -\sigma - \frac{2(1 - AR)\,\mathcal{P}}{1 - R^2}, \qquad a \cdot n = -\frac{\sigma}{2} - \frac{A^2\mathcal{P}}{1 + AR},$$

and $D \cdot (D - 2a) < 0$ if and only if $(1 - AR)\mathcal{P} < -\frac{\sigma}{2}(1 - R^2)$ on the zone. The extremal profile $\mathcal{P} = -\frac{\sigma}{2}\frac{1-R^2}{1-AR}$ corresponds to $\xi = \frac{\sigma}{2}F_1'(R)$ with F_1 as in Example 3.6, whose fronts are at

$$\xi_+ = -\frac{\sigma}{1 - A}, \qquad \xi_- = \frac{\sigma}{1 + A}.$$

Thus, formally, the upper and lower boundaries of the mixing zone may move away from the central interface with normal speeds up to $-\sigma/(1 - A)$ and $-\sigma/(1 + A)$, and the width may grow at any rate below $-2\sigma/(1 - A^2)$. For the flat interface ($\sigma = -2$) this is Example 3.6 with

$\theta \to 1$. By analogy with [FS18, Section 5], we expect that piecewise constant subsolutions with finitely many intermediate densities approximate these rates, and that the proof of Theorem 1.1 extends to them with the same use of $(I - 2A\mathcal{K}(z_0))^{-1}$.

6.3. **The role of the Saffman–Taylor function.** The proof shows that, at the level of subsolutions, the Saffman–Taylor function plays three roles at once: it is the density of the source part of the Darcy field (so that the field of the subsolution is linear in σ), it bounds the admissible opening rate, and it is the unknown of the only operator that has to be inverted. The gravitational and viscous contributions to the instability enter only through $\sigma = -2(n_2 + AU)$, which is why the mixing zone of Example 3.6 is asymmetric: the relaxed flux is weighted by the mobility $1/(1 - A\rho)$, in the spirit of the Koval model for miscible viscous fingering [Kov63, MO05].

6.4. **Open problems.**

(1) *Partially unstable data.* If σ_0 changes sign, mixing should only occur where $\sigma_0 < 0$, as in [CFM22] for $A = 0$. This requires κ to vanish at the boundary of the unstable region, which is excluded by the lower bound $\min \kappa > 0$ used in Lemma 4.5.
(2) *Longer times.* The time expansion only gives a short existence time. Continuing the construction requires an evolution equation for the central interface, as in [CCF21, ACF21]; when $A \neq 0$ it should involve the averaged version of $(I - 2A\mathcal{K})^{-1}$ along the mixing zone.
(3) *Regularity.* We assumed $z_0 \in C^{6,\gamma}$, which is far from the $C^{3,\alpha}$ of [FS18] for $A = 0$. The losses come from the crude bookkeeping in Lemma 5.4.
(4) *Selection.* For $A = 0$, Otto's relaxation selects the maximal mixing rate [Ott99, Sze12]; it would be interesting to know whether a similar selection principle singles out the rates of the previous subsection when $A \neq 0$, see [Men22] for the flat case.

Appendix A. Periodic Cauchy integrals

It is convenient to work with complex densities. For $z \in \mathcal{Z}^{1,\gamma}$ and $G \in C^{0,\gamma}(\mathbb{T};\mathbb{C})$ set

$$\mathfrak{c}[z;G](x) := \frac{1}{4\pi i}\int_{\mathbb{T}} G(\beta)\,\partial_\beta z(\beta)\cot\frac{z(\beta)-x}{2}\,\mathrm{d}\beta, \qquad x \in \mathbb{C}\setminus\Gamma,$$

so that $\mathcal{C}[z;f]^\flat = \mathfrak{c}[z;f^\flat]$.

A.1. **Reduction to closed curves.** With $w = e^{ix}$ and $\omega = e^{i\zeta}$, a direct computation gives

$$\tfrac{1}{2}\cot\tfrac{\zeta-x}{2}\,\mathrm{d}\zeta = \frac{\mathrm{d}\omega}{\omega - w} - \frac{i}{2}\,\mathrm{d}\zeta.$$

The map $x \mapsto e^{ix}$ sends the cylinder $\mathbb{T}\times\mathbb{R}$ to $\mathbb{C}\setminus\{0\}$, Γ to a closed Jordan curve γ of class $C^{k,\gamma}$ winding once around the origin, $\Omega^+(z)$ to the punctured interior of γ and $\Omega^-(z)$ to its exterior. Hence $\mathfrak{c}[z;G]$ is the classical Cauchy integral $\frac{1}{2\pi i}\int_\gamma \frac{\tilde G(\omega)\,\mathrm{d}\omega}{\omega - w}$ of the transported density $\tilde G$, minus the constant $\frac{1}{4\pi}\int_{\mathbb{T}} G\partial_\beta z\,\mathrm{d}\beta$. The classical Plemelj–Privalov theorem for closed $C^{1,\gamma}$ curves [Mus53, Chapter 2], [Kre14, Chapter 7] then gives Lemma 2.3(b): $\mathfrak{c}[z;G]$ extends to a $C^{0,\gamma}$ function on each closed region, with norm bounded by $C\|G\|_{C^{0,\gamma}}$, where C depends only on γ, $\delta(z)$ and $\|z\|_{\mathcal{Z}^{1,\gamma}}$; its boundary values are given by the Plemelj formulas $\mathfrak{c}^\pm = \pm\frac{1}{2}G + \text{p. v. }\mathfrak{c}$. The value at $w = 0$ corresponds to the limit $x_2 \to +\infty$.

A.2. **Derivative identities.**

Lemma A.1. *Let $z \in \mathcal{Z}^{2,\gamma}$ and $G \in C^{1,\gamma}(\mathbb{T};\mathbb{C})$. Then, in $\mathbb{C}\setminus\Gamma$,*

$$\partial_x\,\mathfrak{c}[z;G] = \mathfrak{c}\Big[z;\frac{\partial_\beta G}{\partial_\beta z}\Big]. \tag{A.1}$$

If $z = z(\beta,t)$ and $G = G(\beta,t)$ are C^1 in t with values in $\mathcal{Z}^{2,\gamma}$ and $C^{1,\gamma}$, then

$$\partial_t\,\mathfrak{c}[z;G](x) = \mathfrak{c}\Big[z;\partial_t G - \frac{\partial_\beta G}{\partial_\beta z}\,\partial_t z\Big](x), \qquad x \notin \Gamma(t). \tag{A.2}$$

Proof. Since $\partial_\beta \cot \frac{z(\beta)-x}{2} = -\frac{1}{2}\csc^2 \frac{z(\beta)-x}{2}\,\partial_\beta z$, we have $\partial_x \cot \frac{z(\beta)-x}{2} = -\partial_\beta \cot \frac{z(\beta)-x}{2}/\partial_\beta z$ and $\partial_t \cot \frac{z(\beta)-x}{2} = \partial_\beta \cot \frac{z(\beta)-x}{2} \cdot \partial_t z/\partial_\beta z$. Integrating by parts in β (there are no boundary terms by periodicity),

$$\partial_x \mathfrak{c} = -\frac{1}{4\pi i}\int G\,\partial_\beta \cot\,\mathrm{d}\beta = \frac{1}{4\pi i}\int \partial_\beta G \cot\,\mathrm{d}\beta,$$

which is (A.1), and

$$\partial_t \mathfrak{c} = \frac{1}{4\pi i}\int \Big[(\partial_t G\,\partial_\beta z + G\,\partial_\beta \partial_t z)\cot - \partial_\beta(G\partial_t z)\cot\Big]\,\mathrm{d}\beta = \frac{1}{4\pi i}\int \big(\partial_t G\,\partial_\beta z - \partial_\beta G\,\partial_t z\big)\cot\,\mathrm{d}\beta,$$

which is (A.2). □

Iterating (A.1) and using the Plemelj–Privalov theorem gives Lemma 2.3(d): if $z \in \mathcal{Z}^{j+1,\gamma}$ and $G \in C^{j,\gamma}$ then all derivatives of order $\le j$ of $\mathfrak{c}[z;G]$ are Cauchy integrals of $C^{0,\gamma}$ densities, hence $C^{0,\gamma}$ up to Γ. If $x = x(t)$ is a C^1 path in $\Omega^\pm(z(t))$, then since $\mathfrak{c}$ is holomorphic,

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\mathfrak{c}[z;G]\big(x(t)\big) = \mathfrak{c}\Big[z;\partial_t G - \frac{\partial_\beta G}{\partial_\beta z}\partial_t z\Big](x) + \mathfrak{c}\Big[z;\frac{\partial_\beta G}{\partial_\beta z}\Big](x)\,\dot{x}, \tag{A.3}$$

and by uniform convergence of the derivatives up to Γ the same formula holds for the boundary values $t \mapsto \mathfrak{c}[z;G]^\pm(\alpha)$ with $x(t) = z(\alpha,t)$.

A.3. **Continuity.**

Lemma A.2. *Let $z_k \to z$ in $\mathcal{Z}^{1,\gamma}$ and $G_k \to G$ in $C^{0,\gamma}(\mathbb{T};\mathbb{C})$. Then $\mathfrak{c}[z_k;G_k]^\pm \to \mathfrak{c}[z;G]^\pm$ uniformly on $\mathbb{T}$, and if $x_k \in \overline{\Omega^+(z_k)}$ with $x_k \to z(\alpha)$, then $\mathfrak{c}[z_k;G_k](x_k) \to \mathfrak{c}[z;G]^+(\alpha)$; similarly from below.*

Proof. By the Plemelj–Privalov theorem the functions $\mathfrak{c}_k := \mathfrak{c}[z_k;G_k]$ are uniformly $C^{0,\gamma}$ on $\overline{\Omega^+(z_k)}$, with a constant L independent of k. Given $\varepsilon > 0$ let $y = z(\alpha) + \varepsilon n(\alpha)$; for k large $y \in \Omega^+(z_k)$ at distance $\ge \varepsilon/2$ from Γ_k, so $\mathfrak{c}_k(y) \to \mathfrak{c}[z;G](y)$ by dominated convergence. Hence

$$\limsup_k\,|\mathfrak{c}_k(x_k) - \mathfrak{c}[z;G]^+(\alpha)| \le \limsup_k\,\big(L|x_k - y|^\gamma + L\varepsilon^\gamma\big) \le 2L\varepsilon^\gamma,$$

and ε is arbitrary. Uniformity in α follows from the same argument and the equicontinuity. □

A.4. **Proof of Lemma 5.4 and of the regularity of G in Lemma 5.5.**

By Lemma 4.1, the curves $z^\pm(\cdot,t)$ are polynomials in t with coefficients in $C^{3,\gamma}$, with $\delta(z^\pm) \ge \frac{1}{2}\delta(z_0)$ on $[0,T_1]$. The normals $n^\pm$, the lengths $|\partial_\alpha z^\pm|$ and the velocities $V^\pm = \partial_t z^\pm \cdot n^\pm$ are therefore smooth in t with values in $C^{2,\gamma}$, and so are the complex densities $G^\pm := (f^\pm)^\flat$ and their time derivatives.

By (4.5), $(v_S^+)^\flat = \mathfrak{c}[z^+;G^+]^- + \mathfrak{c}[z^-;G^-]\circ z^+$, where $z^+(\alpha,t) \in \Omega^+(z^-(\cdot,t))$ for $t > 0$, and similarly for v_S^-. Applying (A.3) once, $\partial_t v_S^\pm$ is a sum of boundary values and evaluations on the other curve of Cauchy integrals with densities

$$G_1^\pm = \partial_t G^\pm - \frac{\partial_\beta G^\pm}{\partial_\beta z^\pm}\partial_t z^\pm, \qquad G_2^\pm = \frac{\partial_\beta G^\pm}{\partial_\beta z^\pm},$$

multiplied by $\partial_t z^\pm$; these densities are bounded in $C^{1,\gamma}$ uniformly in t. Applying (A.3) once more, $\partial_t^2 v_S^\pm$ involves Cauchy integrals of densities bounded in $C^{0,\gamma}$, evaluated on $\Gamma^\pm(t)$ from the side of the strip, and these are bounded uniformly in $t \in (0,T_1]$ by the Plemelj–Privalov theorem. Since $F^\pm$ are polynomials in $v_S^\pm \cdot n^\pm$, $V^\pm$, $n_2^\pm$ and $|\partial_\alpha z^\pm|$, we obtain that $\partial_t F^\pm$ and $\partial_t^2 F^\pm$ are bounded in $C^0(\mathbb{T})$ on $(0,T_1]$. By Lemma A.2 applied to the densities appearing in $F^\pm$ and $\partial_t F^\pm$, these functions extend continuously to $t = 0$, which proves the first part of Lemma 5.4.

For (5.4), let $x = X(\alpha,\lambda,t)$, so that $|x - z^\pm(\alpha,t)| \le 2t\kappa(\alpha)$. The uniform Hölder bounds on $\overline{\Omega^-(z^+)}$ and $\overline{\Omega^+(z^-)}$ give $|v(x,t) - v_S^+(\alpha,t)| \le Ct^\gamma$, and the Lipschitz bound in time just proved gives $|v_S^+(\alpha,t) - v_S^{(0)}(\alpha)| \le Ct$.

Finally, $G = \partial_t M(\cdot,0)$ for $a = 0$, $h = 0$ is obtained by letting $t \to 0$ in the formula for $\partial_t M$. In the limit all Cauchy integrals become boundary values on $z_0 \in \mathcal{Z}^{6,\gamma}$, and the densities

involve $n_0 \in C^{5,\gamma}$, $\partial_t z^\pm(\cdot,0) = (U_0 \pm c^\pm)n_0 \in C^{4,\gamma}$, their first α-derivatives, and $\partial_t^2 z^\pm(\cdot,0) = 0$. Thus $G^\pm(\cdot,0) \in C^{4,\gamma}$, $G_1^\pm(\cdot,0), G_2^\pm(\cdot,0) \in C^{3,\gamma}$, and by Lemma 2.3(d) the boundary values are in $C^{3,\gamma}$. The remaining factors ($\partial_t z^\pm(\cdot,0)$, n_0, and the time derivatives at $t=0$ of $n^\pm$, $V^\pm$ and $|\partial_\alpha z^\pm|$, which involve $\partial_\alpha \partial_t z^\pm(\cdot,0)$) belong to $C^{3,\gamma}$. Hence $G \in C^{3,\gamma}(\mathbb{T})$. □

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